

Lecture 12:

Secure Computation Technologies

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13/05/2025



Universiteit
Leiden

Bij ons leer je de wereld kennen

Secure Computation Technologies

- Reminder of Homomorphism in RSA
- Why Secure Computation?
- Securing Data in Use – PETs (Privacy Enhancing Technologies)
 - Differential Privacy
 - Data Anonymization
 - Cryptographic Techniques
 - Homomorphic Encryption
 - Multiparty Computation (MPC)
- A closer look at MPC
- Concrete Application Scenarios

Privacy Enhancing Technologies



Homomorphism in RSA

Given: $c_1 = m_1^e \bmod n$ and $c_2 = m_2^e \bmod n$

Unpadded RSA allows us to compute:

$$c_1 * c_2 = (m_1^e \bmod n) * (m_2^e \bmod n) = (m_1 * m_2)^e \bmod n ,$$

which then decrypts to the product of the two plaintexts:

$$(c_1 * c_2)^d \bmod n = m_1 * m_2$$

- Malleability... allows for attacks
- But, also.... for Secure Computation schemes

Why Secure Computation?

The three states of data

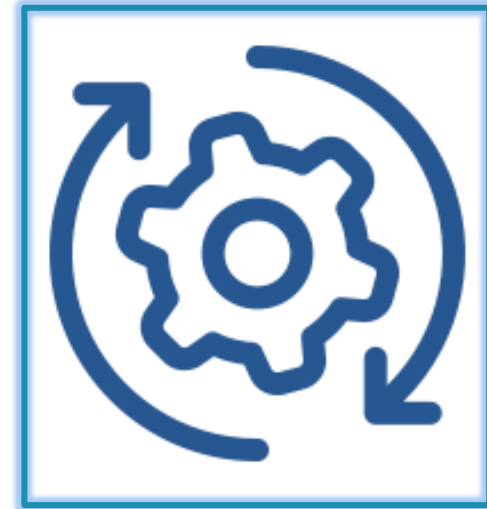
Data in Transit



Data at Rest



Data in Use



The three states of data

Traditional Cryptography

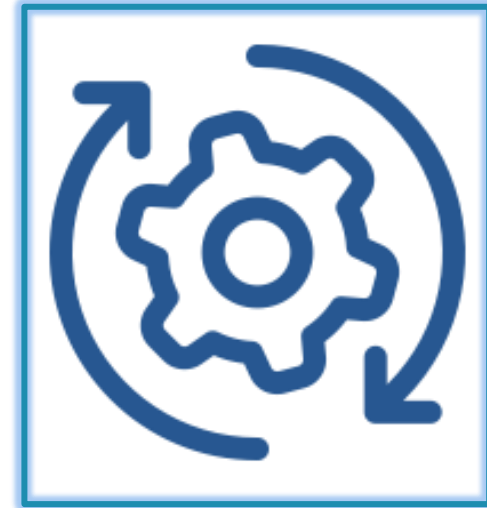
Data in Transit



Data at Rest



Data in Use



The three states of data

Traditional Cryptography

Data in Transit

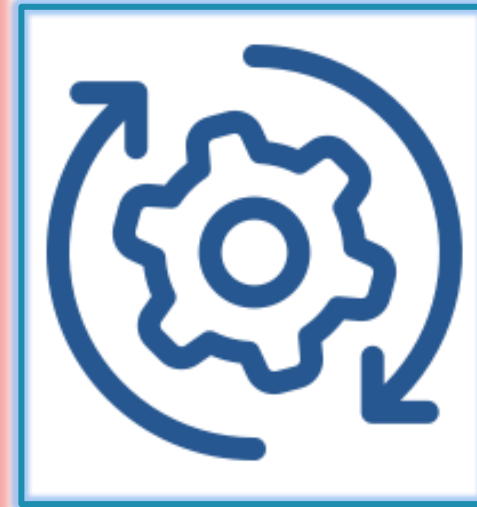


Data at Rest



Secure Computation

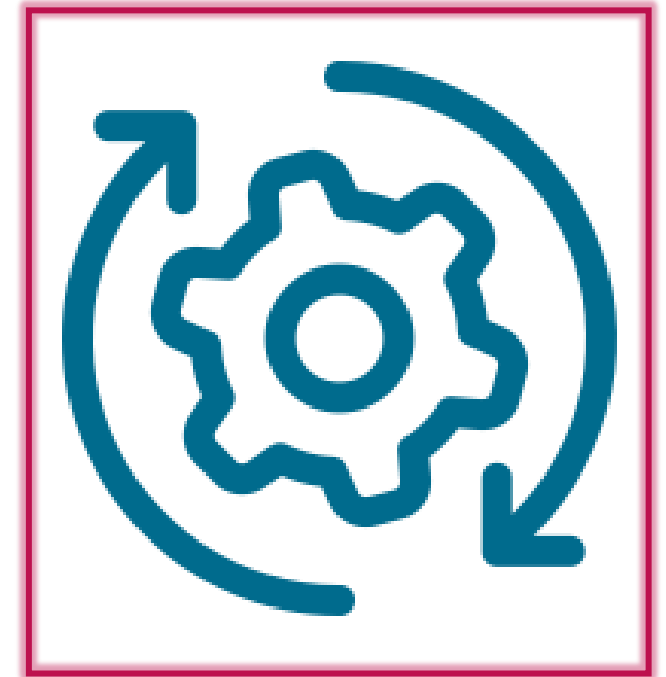
Data in Use



Data in Use



Data in Use

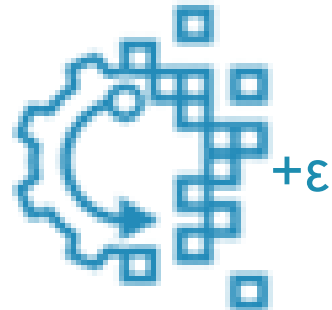


“How to allow the collection and purposeful processing of private data, without compromising individual privacy?”

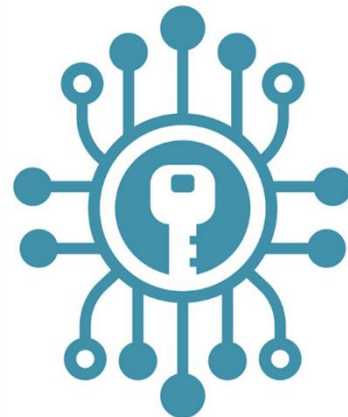
Securing Data in Use – PETs



Data Anonymization

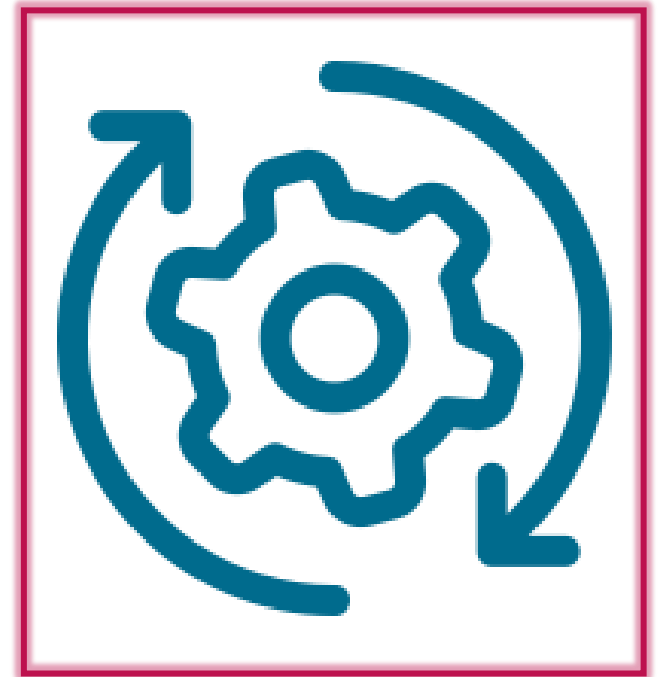


Differential Privacy

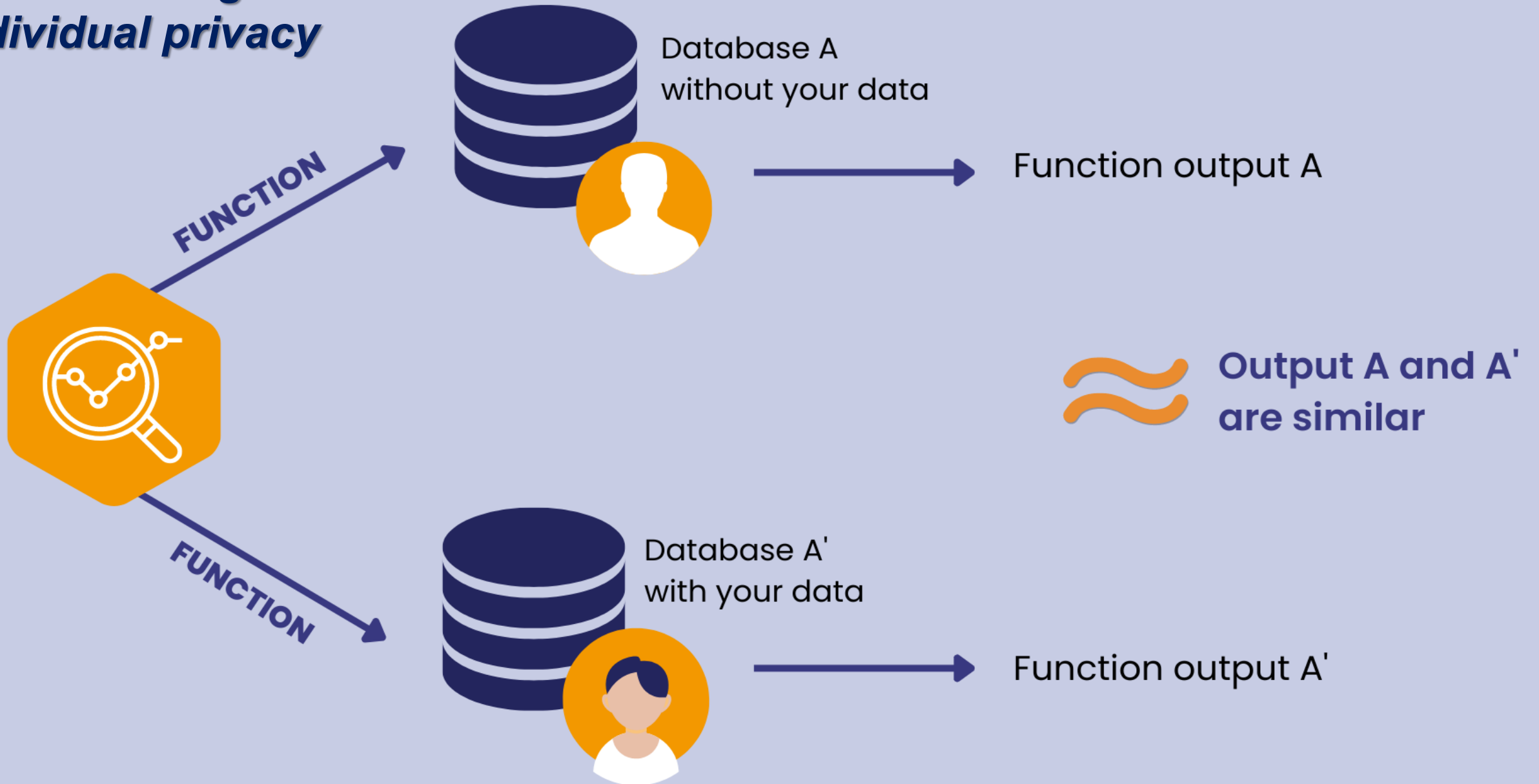


Cryptographic Techniques
for Secure Computation

Data in Use



DP – Ensuring individual privacy



DP – Ensuring individual privacy

FUNCTION



SENSITIVITY



Determine the
function sensitivity



Choose a noise
addition mechanism

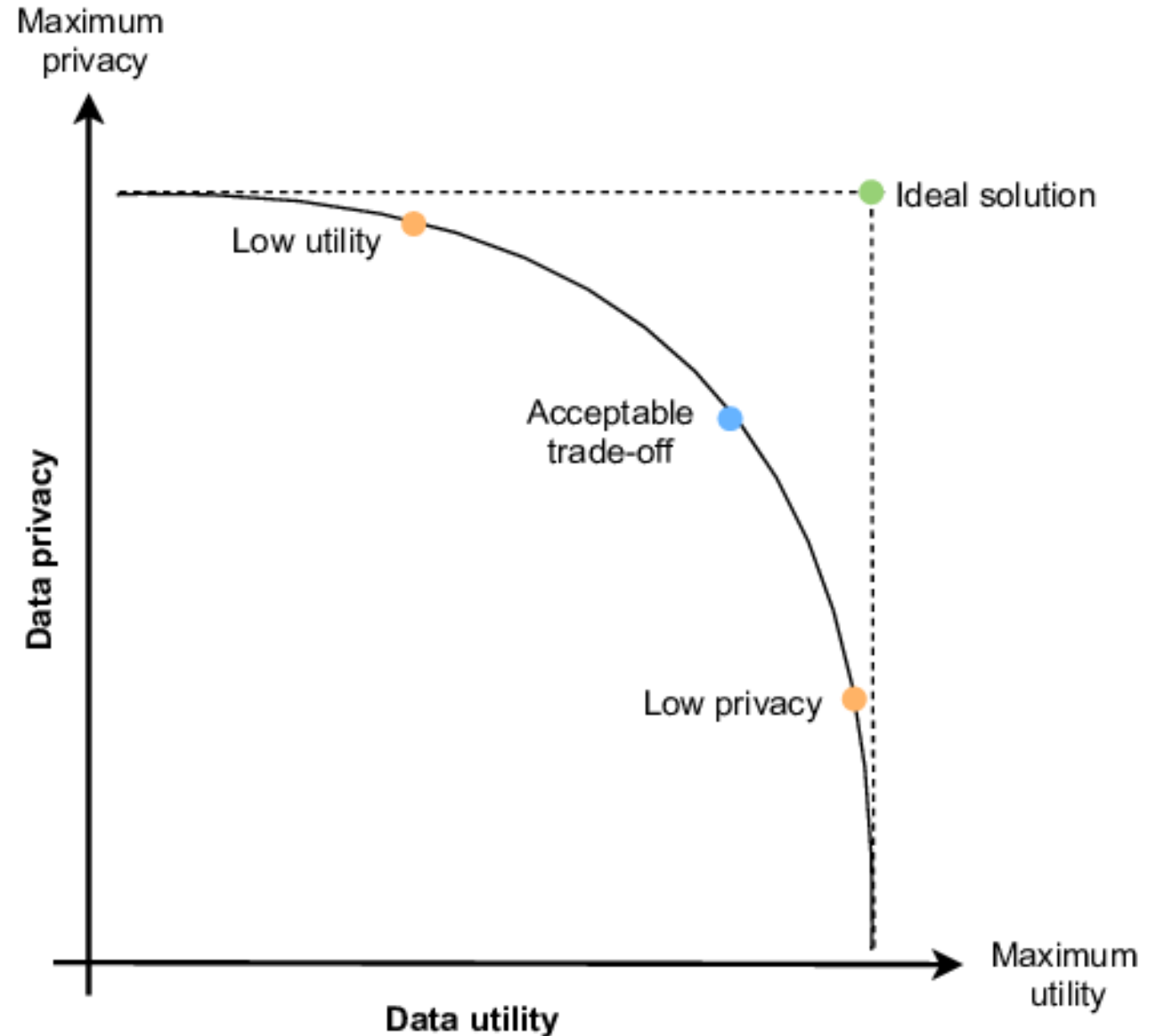


DATA WITH NOISE



DP – The Penalties

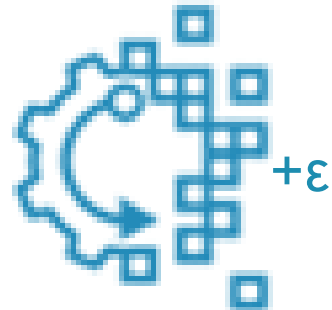
- Inherent information leakage, called privacy budget in DP terms.
- Inherent tradeoff between data privacy and data utility



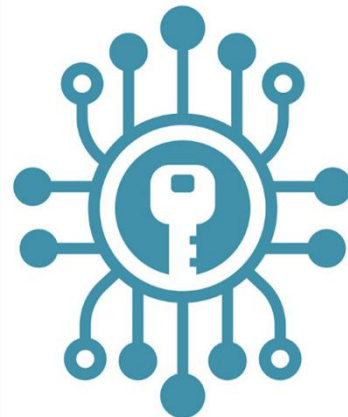
Securing Data in Use – PETs



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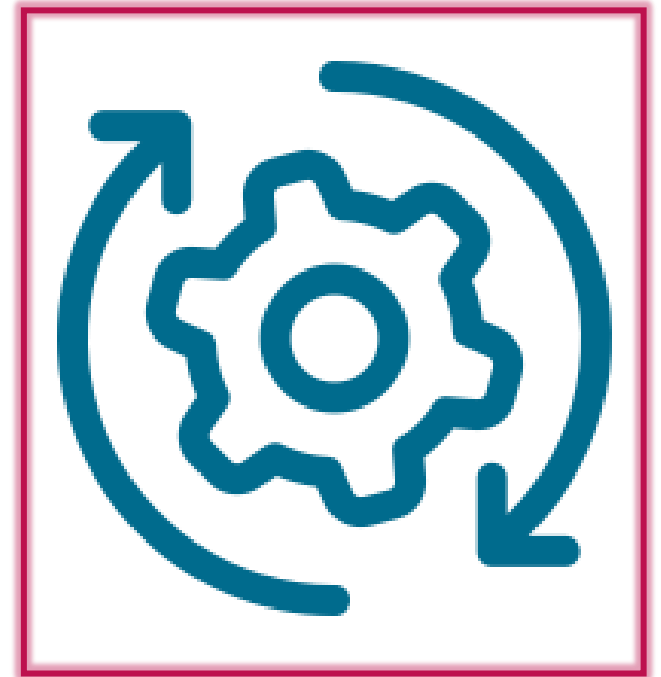


Differential Privacy

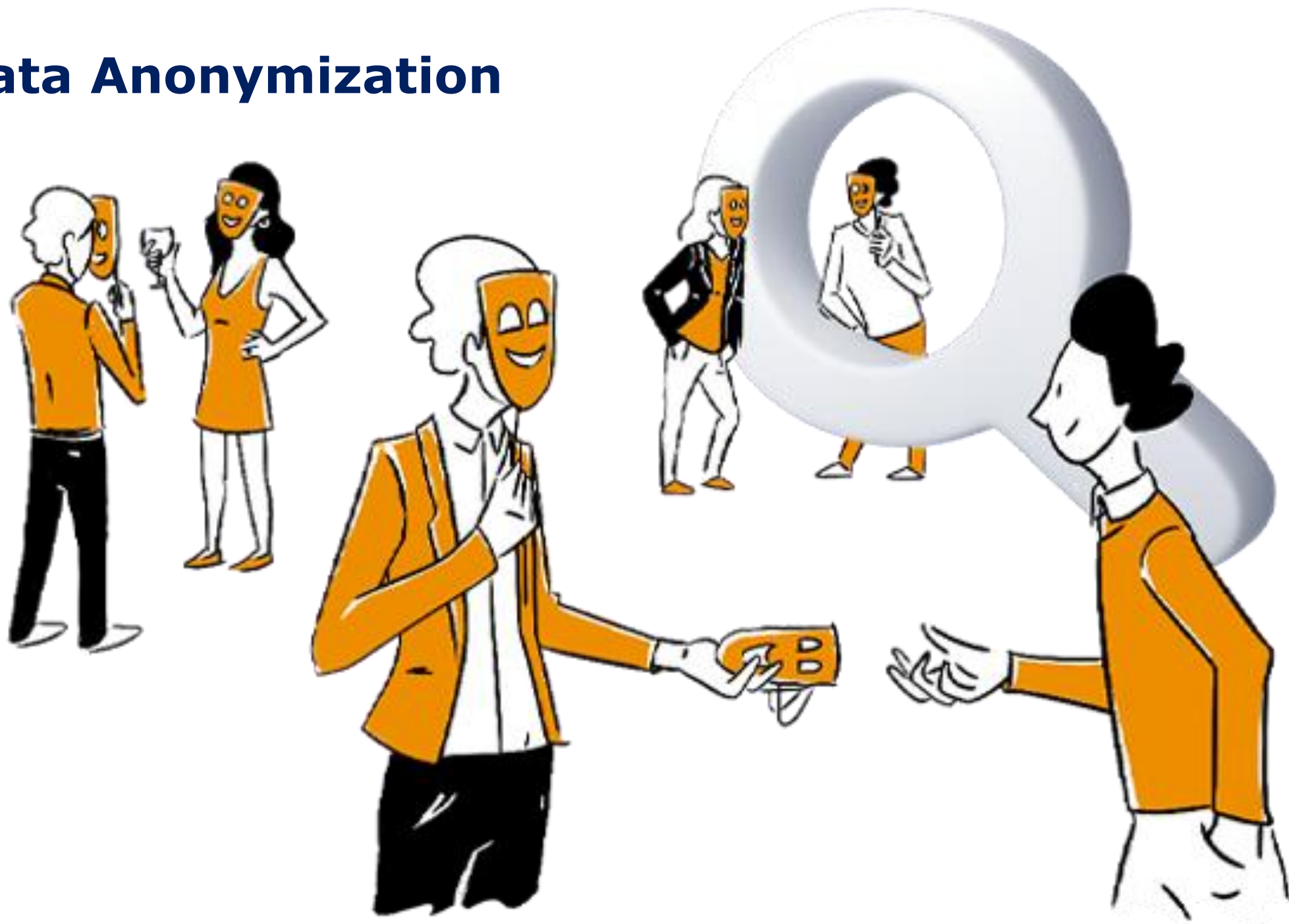


Cryptographic Techniques
for Secure Computation

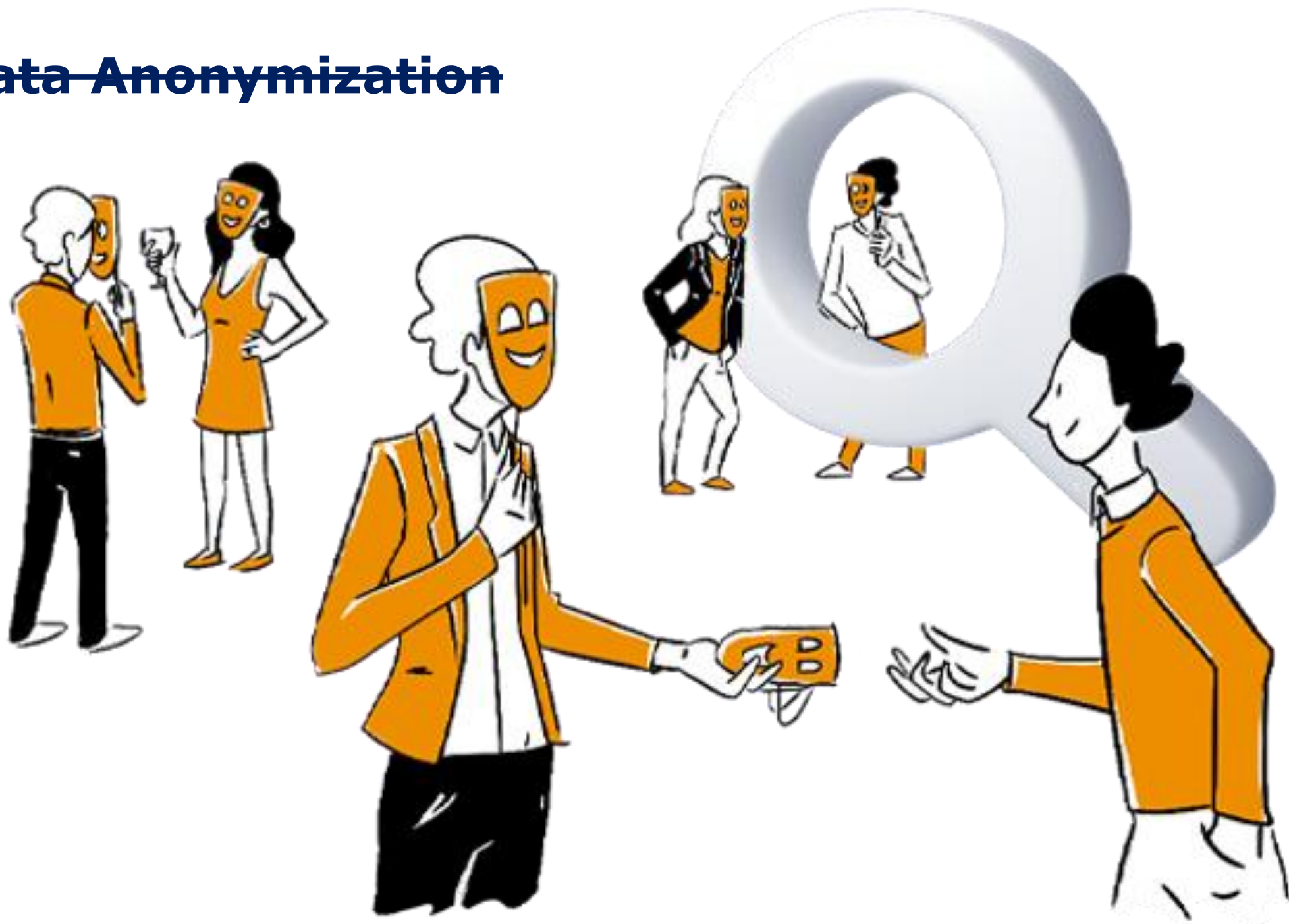
Data in Use



Data Anonymization



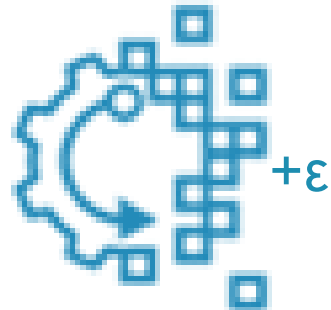
~~Data Anonymization~~



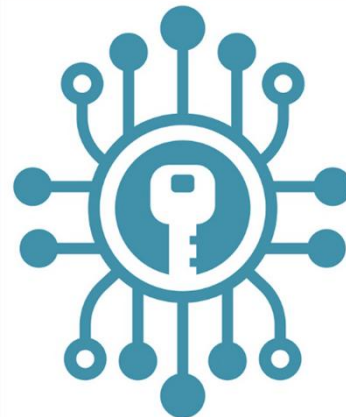
Securing Data in Use – PETs



Data Anonymization



Differential Privacy



Cryptographic Techniques
for Secure Computation

Data in Use



Homomorphic Encryption



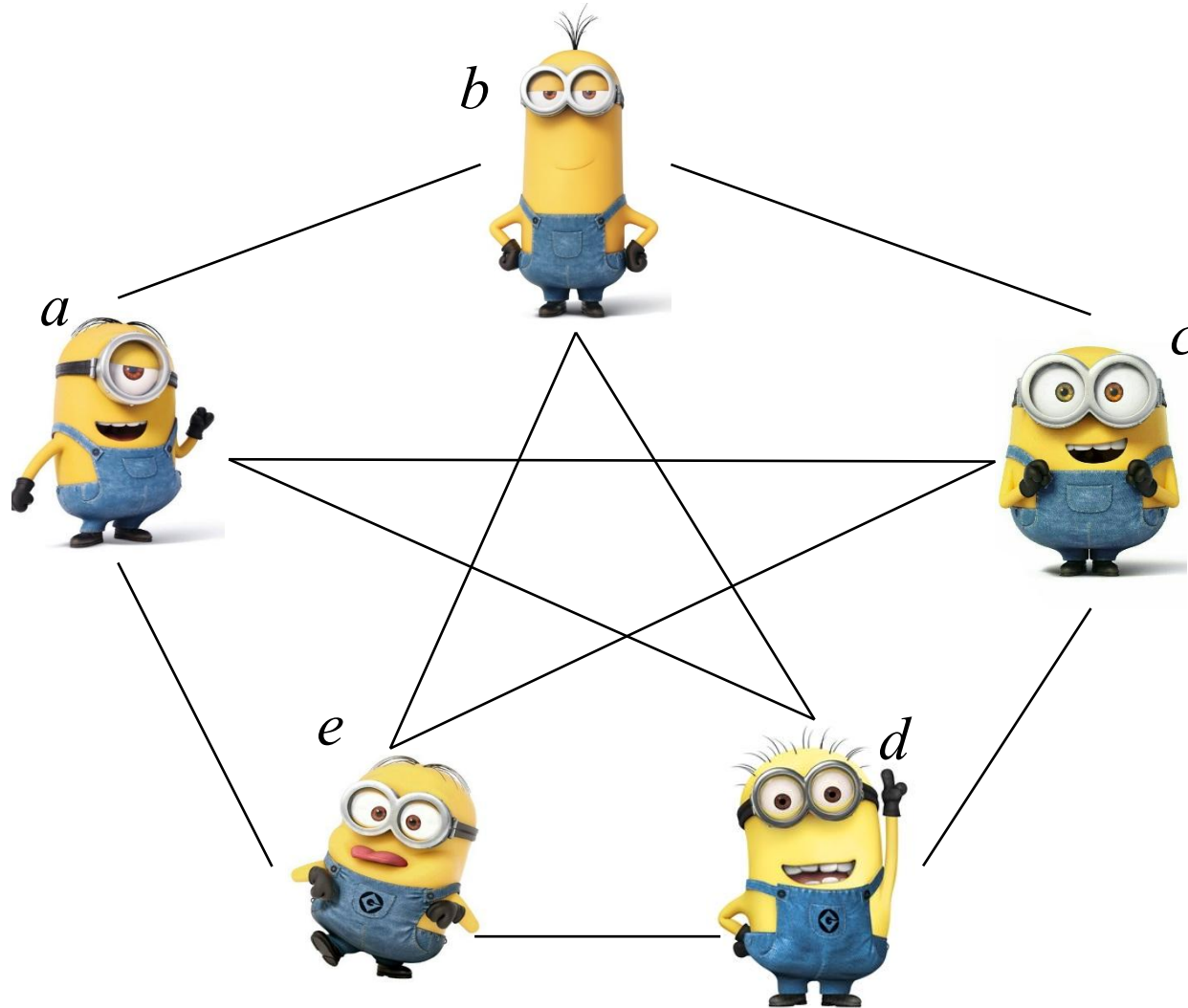
Types of Homomorphic Encryption

HE Type	Type of Operation	Number of Operations
Partially Homomorphic Encryption (PHE)	One (Addition OR Multiplication)	Unlimited
Somewhat Homomorphic Encryption (SHE)	Two (Addition AND Multiplication)	Limited
Fully Homomorphic Encryption (FHE)	Two (Addition AND Multiplication)	Unlimited

Homomorphic Encryption Context

- Outsourcing the computation to ***one single party*** (server)
- Inherently ***trust*** the server to perform the computation correctly and honestly
- ***One round*** of communication: Query – Response
- ***Communication and computational cost*** is considerably larger than plaintext computations

Multiparty Computation – MPC



→ Securely compute $f(a, b, c, d, e)$.

“Two millionaires wish to know who is richer; however, they do not want to find out inadvertently any additional information about each other’s wealth. How can they carry out such a conversation?”

-Andrew Yao, 1982

Multiparty Computation Context

- *A minimum of two parties* (servers) is required to perform computation
- *Trust* is distributed and there are various security models one can consider
- *Multiple rounds* of communication required in an interactive protocol
- *Communication and computational cost* is considerably larger than plaintext computations (yet *more efficient* than HE solutions!)

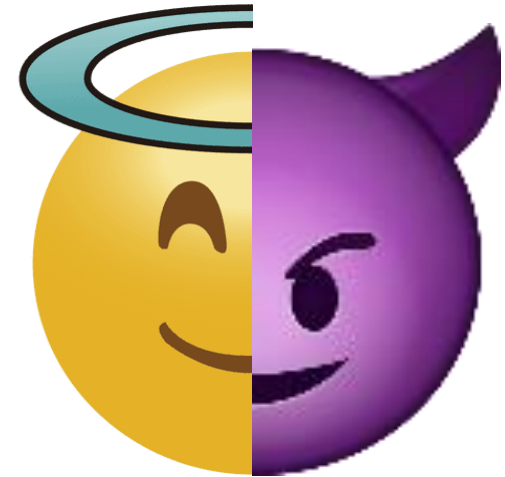
A closer look at MPC

Security Models

- Active Security

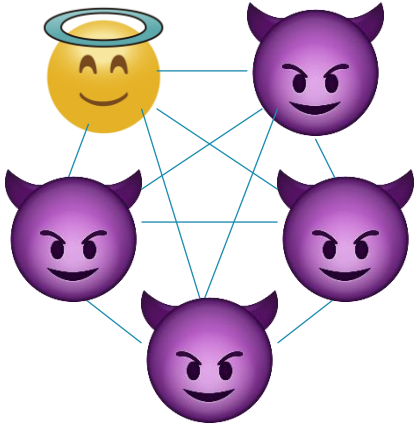
vs.

- Passive Security



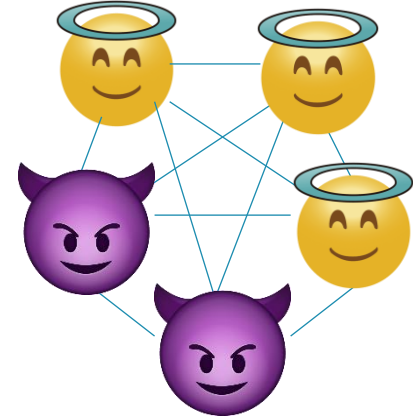
Corruption Thresholds

- Dishonest Majority



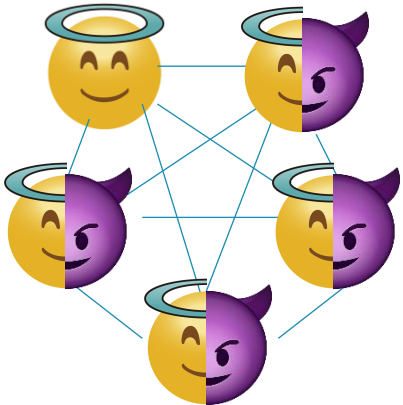
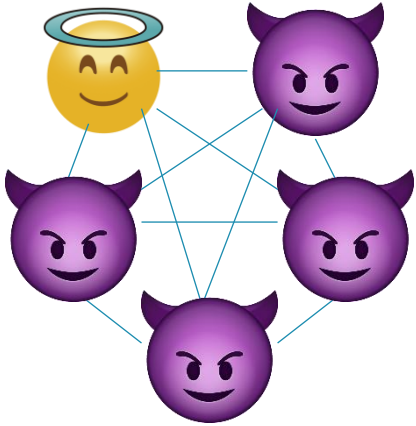
vs.

- Honest Majority



Corruption Thresholds

- Dishonest Majority



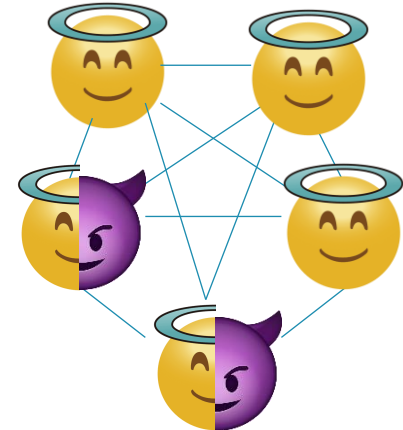
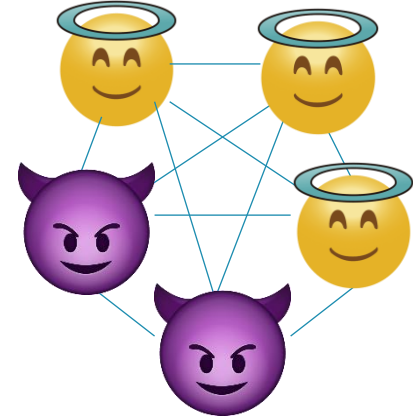
vs.

in the active model

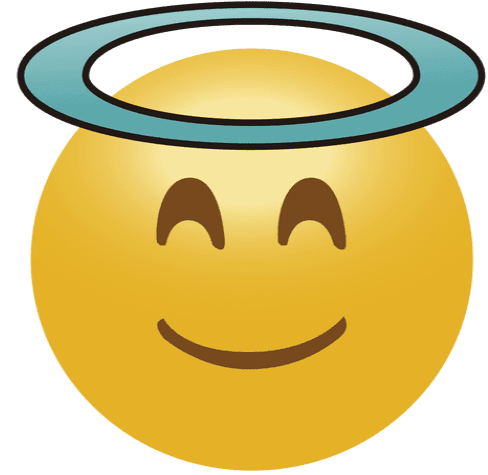
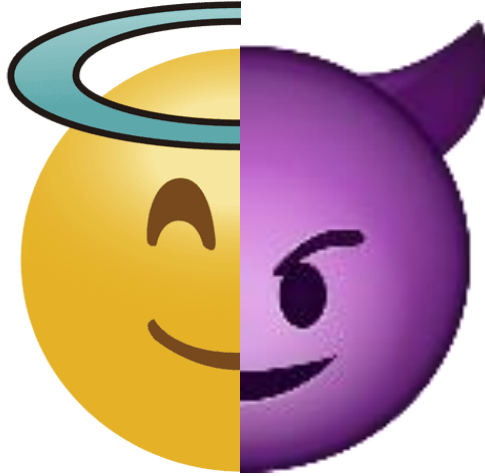
OR

in the passive model

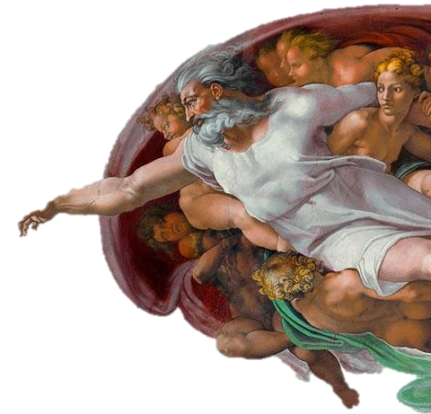
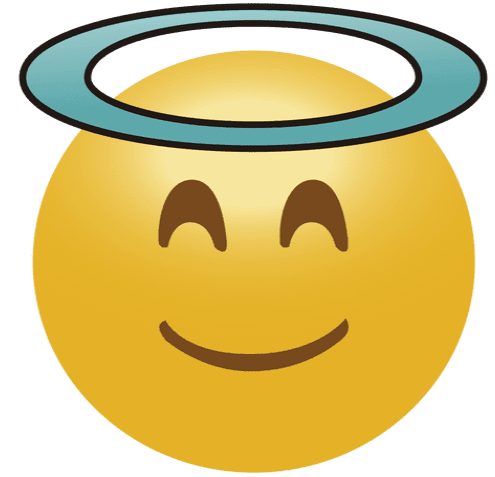
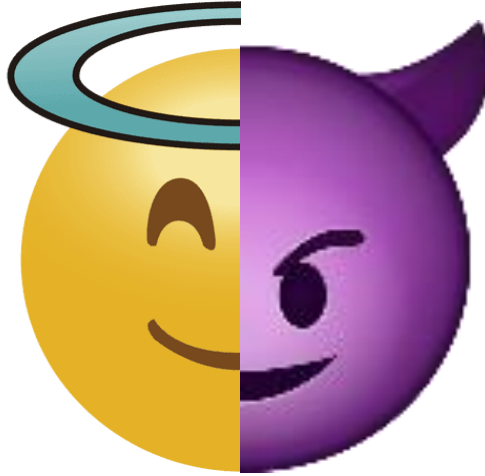
- Honest Majority



Types of Parties



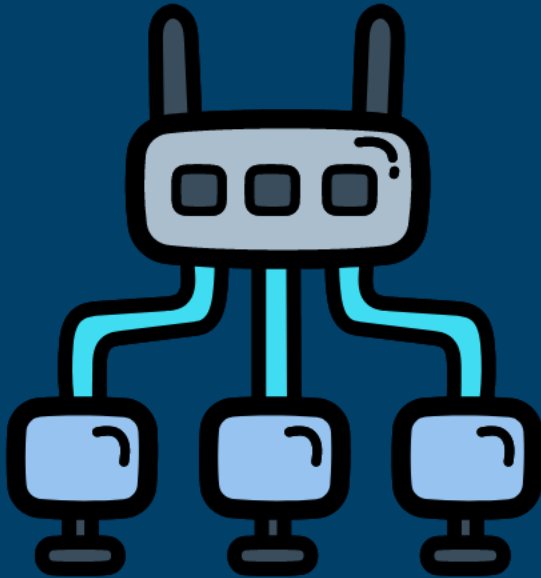
Types of Parties



Computational Power of Parties



Available Network Setting/Resources

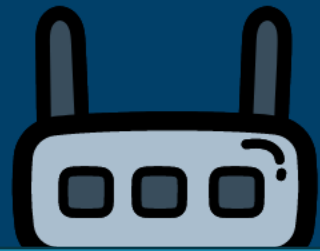


LAN



WAN

Available Network Setting/Resources



Minimum Data Transfer

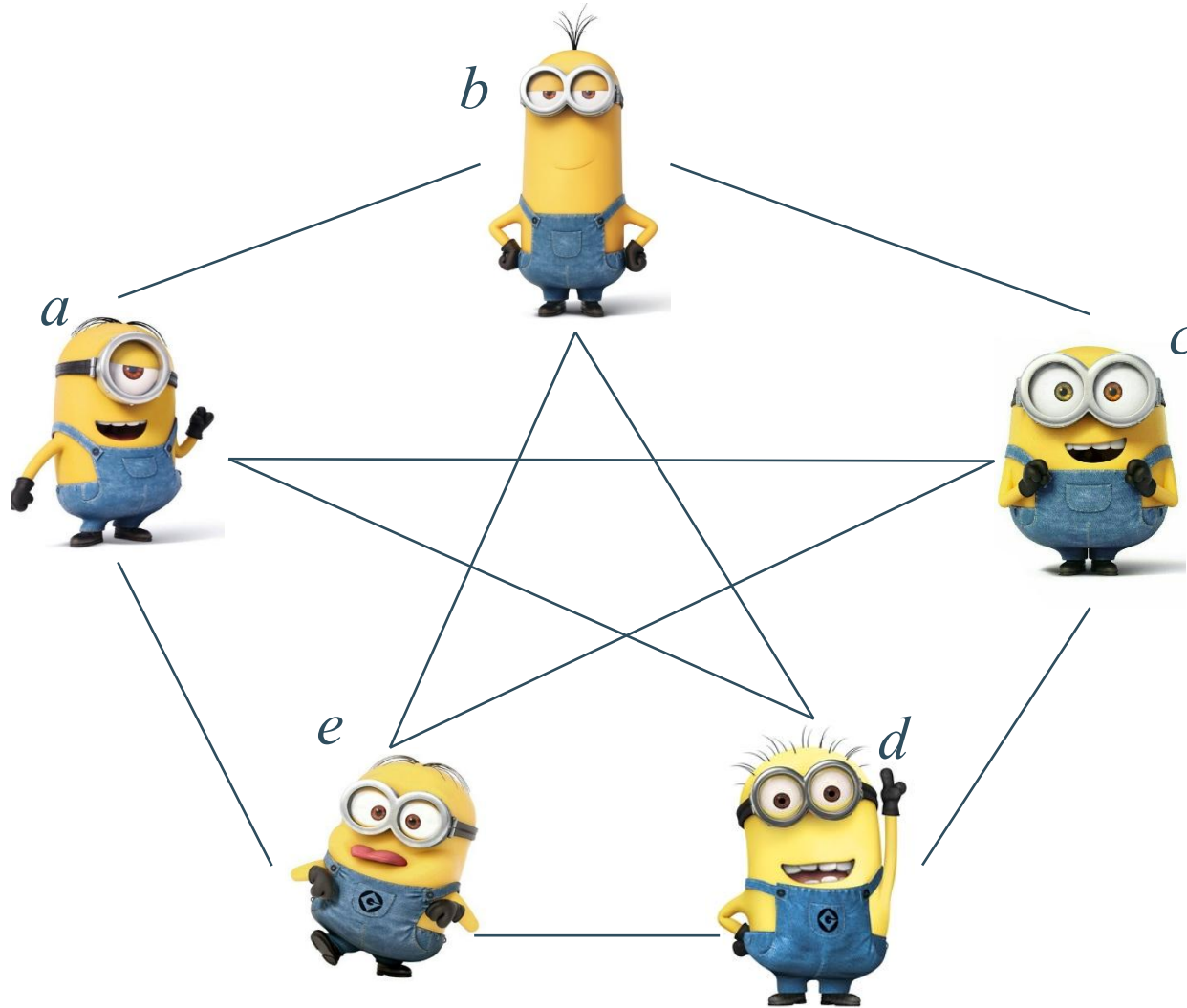
LAN



Minimum Number of Rounds

WAN

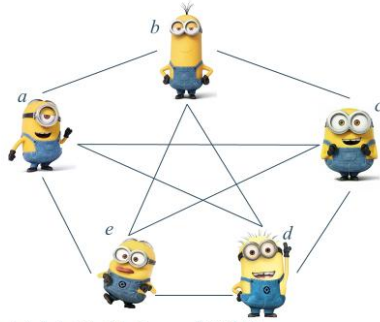
Secret Sharing-based MPC (e.g., additive)



$$\text{Secret } x = a + b + c + d + e \pmod{p}$$

Available Network Setting/Resources

Secret Sharing Based MPC (e.g., additive)



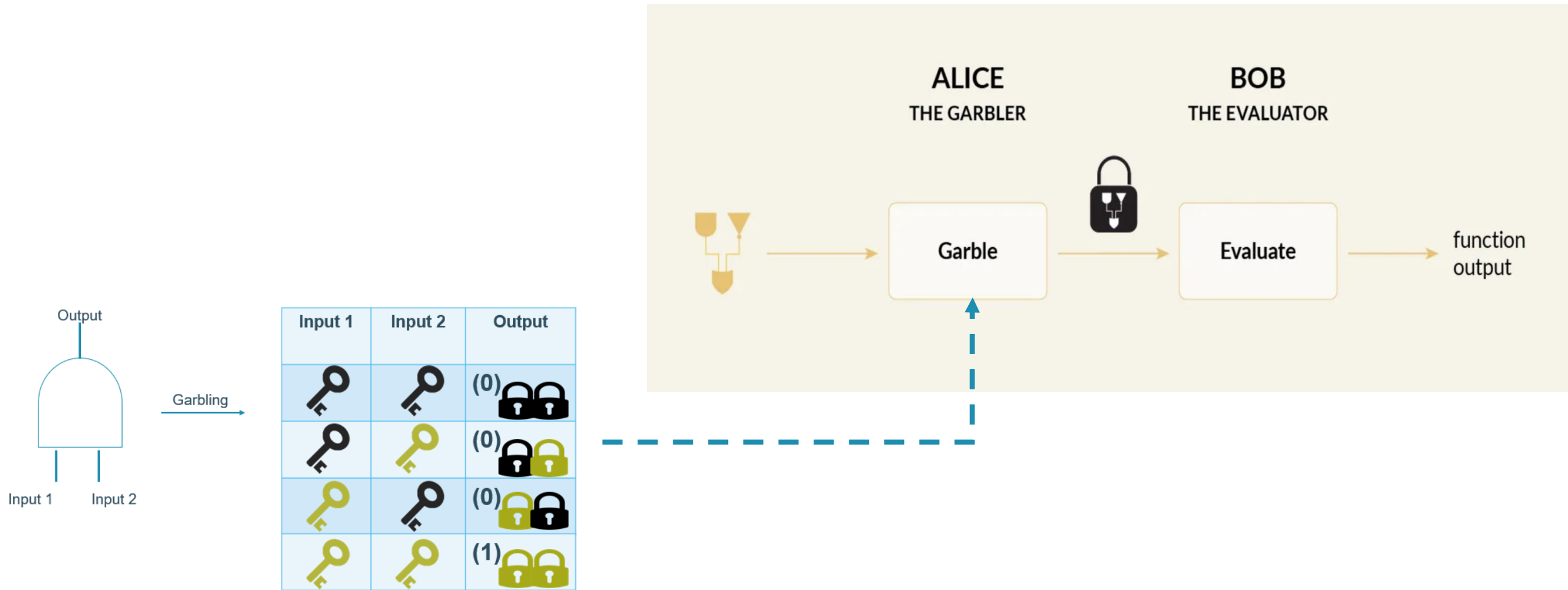
Secret $x = a + b + c + d + e \pmod{p}$

LAN

Minimum Number of Rounds

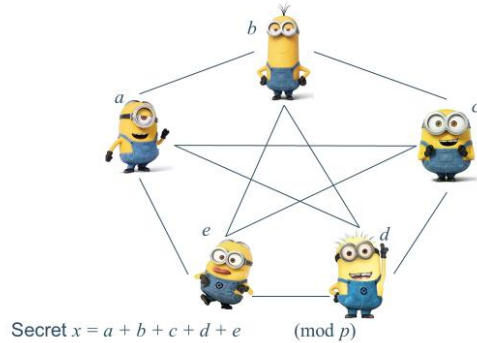
WAN

Garbling Based (MPC)



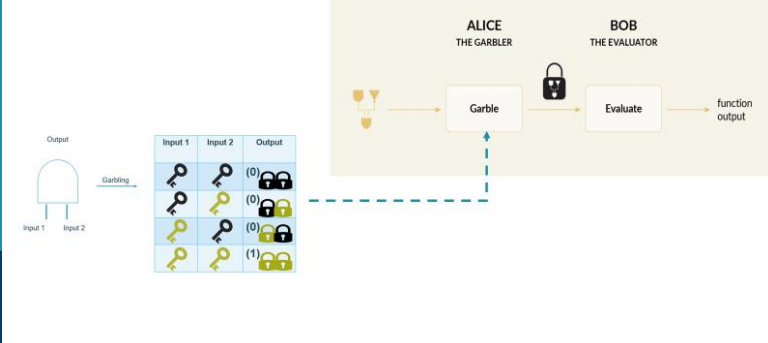
Available Network Setting/Resources

Secret Sharing Based MPC (e.g., additive)



LAN

Garbling Based (MPC)



WAN

The Preprocessing Model

Offline Phase

- Data independent
- Any time prior to protocol execution
- Prepare (random) material for input sharing and multiplication computations

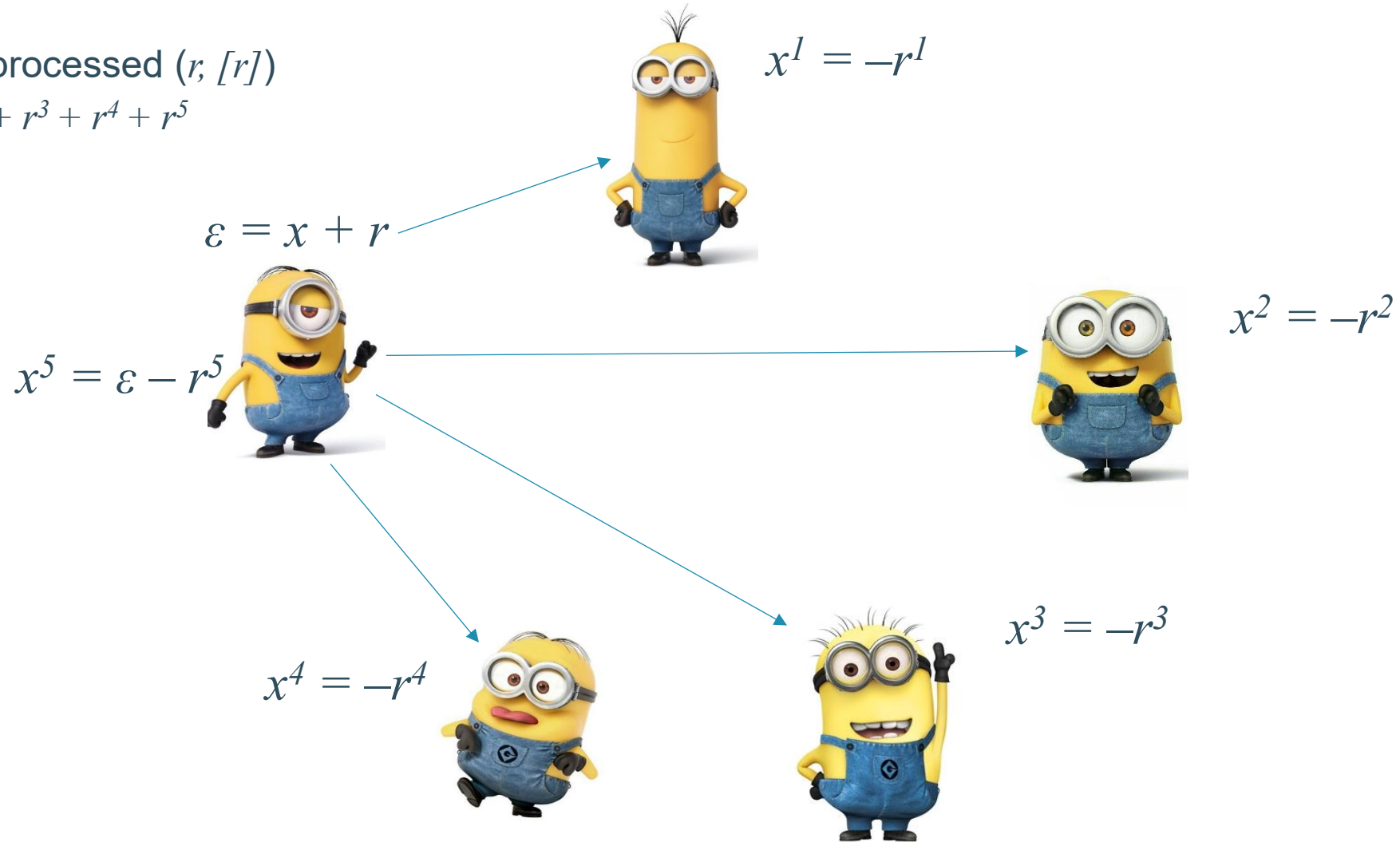
Online Phase

- Data dependent
- Synchronous protocol execution with all parties online
- Consume preprocessed material to (more efficiently) complete the computation at hand

Share Input x (online phase)

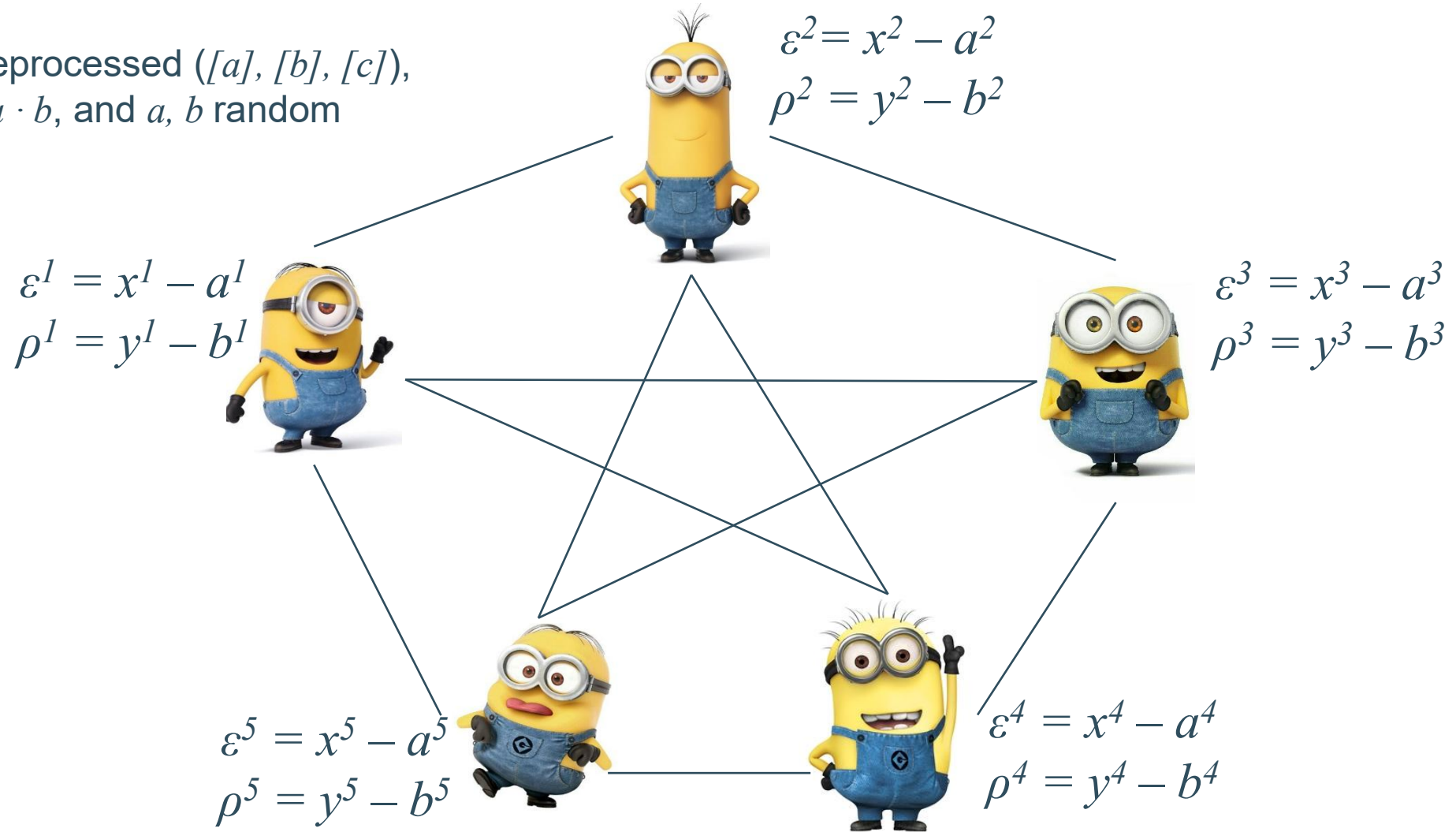
Using preprocessed $(r, [r])$

$$r = r^1 + r^2 + r^3 + r^4 + r^5$$



Compute Product $z = x \cdot y$ (online phase) 1/3

Using preprocessed $([a], [b], [c])$,
with $c = a \cdot b$, and a, b random



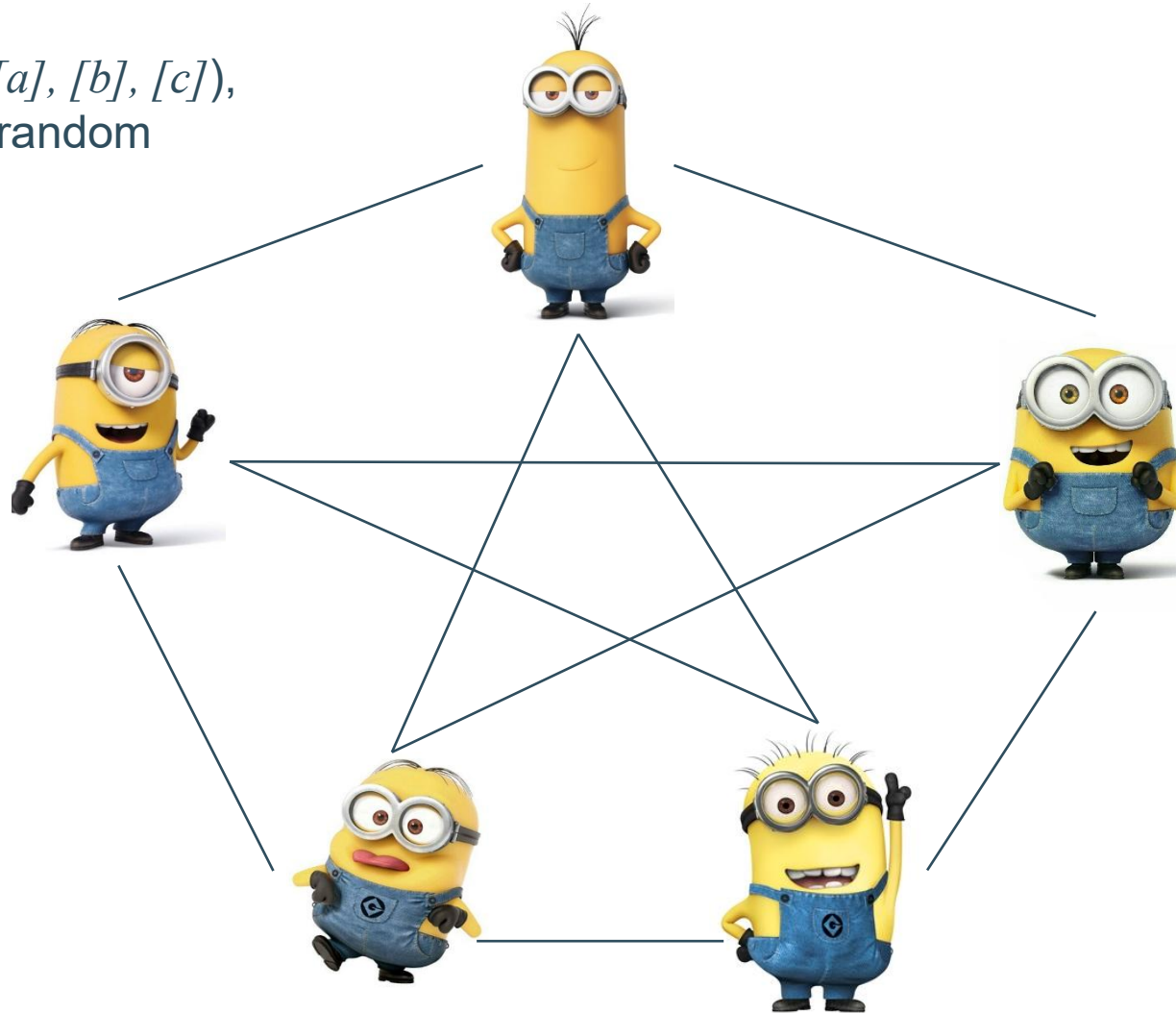
Compute Product $z = x \cdot y$ (online phase) 2/3

Using preprocessed $([a], [b], [c])$,
with $c = a \cdot b$, and a, b random

Open

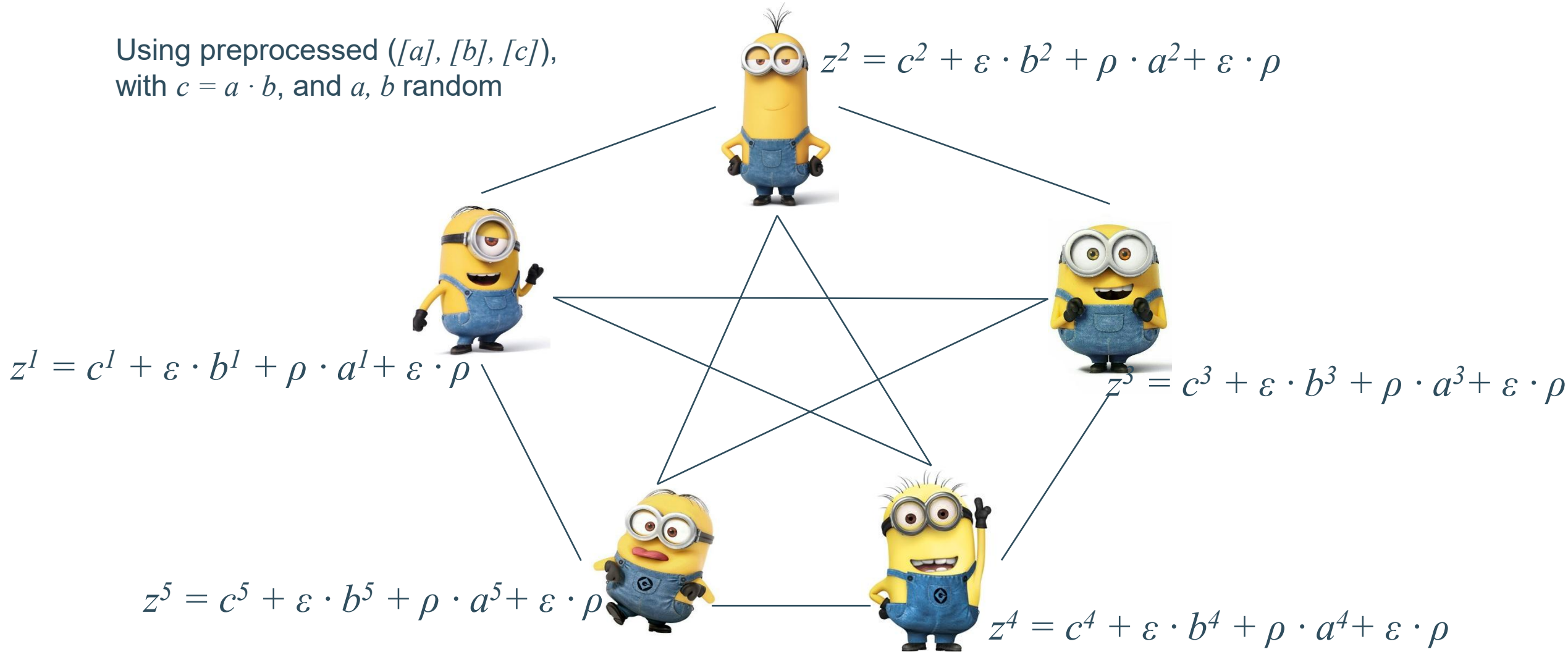
$$\varepsilon = x - a$$

$$\rho = y - b$$



Compute Product $z = x \cdot y$ (online phase) 3/3

Using preprocessed $([a], [b], [c])$,
with $c = a \cdot b$, and a, b random



Set $[z] = [c] + \varepsilon \cdot [b] + \rho \cdot [a] + \varepsilon \cdot \rho$

The tradeoffs



The tradeoffs



MPC Applications

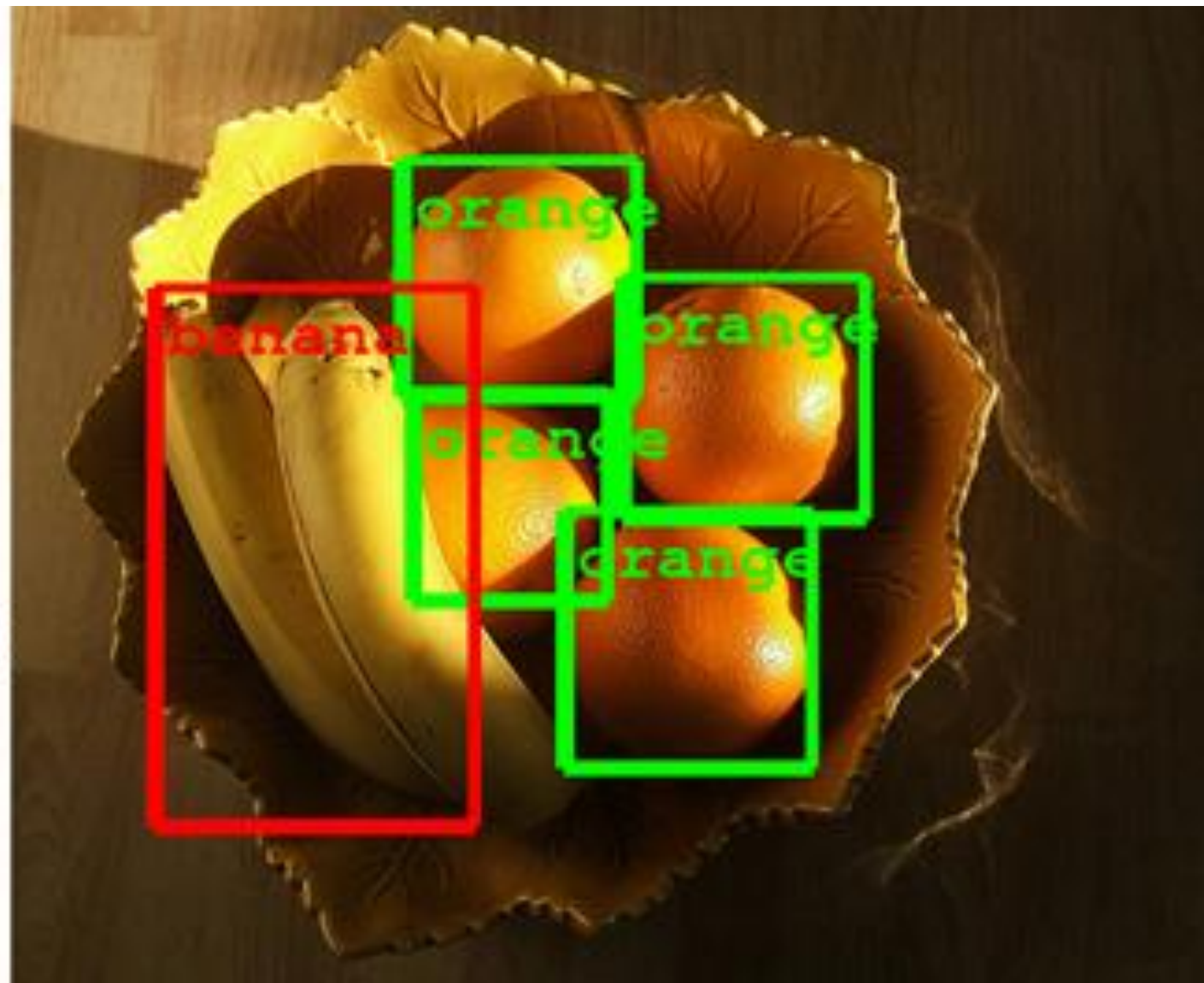
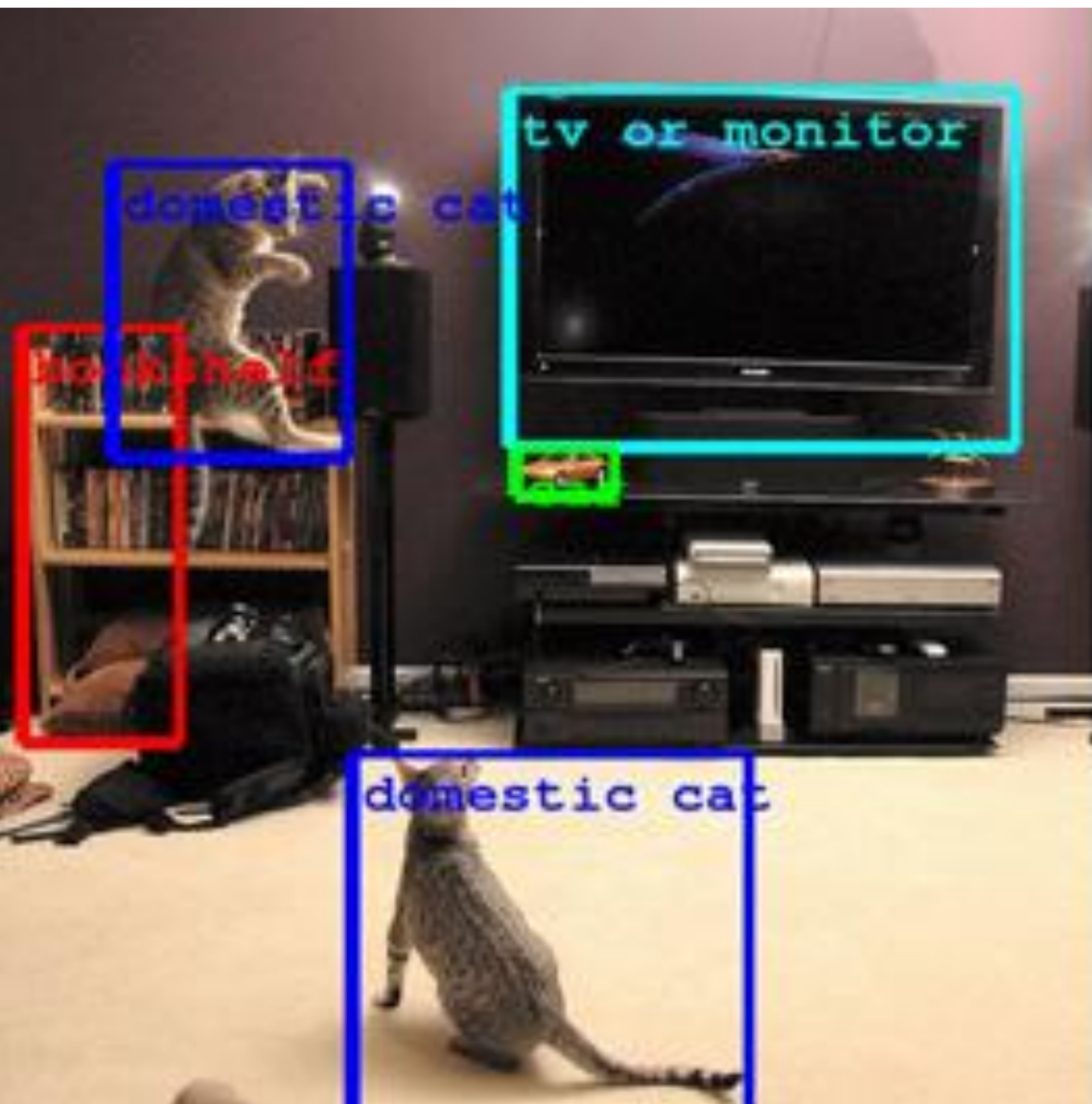
- Auctions
- e-Voting
- Statistical Analysis and Collaborative Analytics
- Fraud Detection
- e-Health and general medical research applications
- Key Management
- Machine Learning

According to Gartner, by 2025, 50% of large organizations will implement privacy-enhancing computation to process data in untrusted environments and multiparty data analytics use cases.

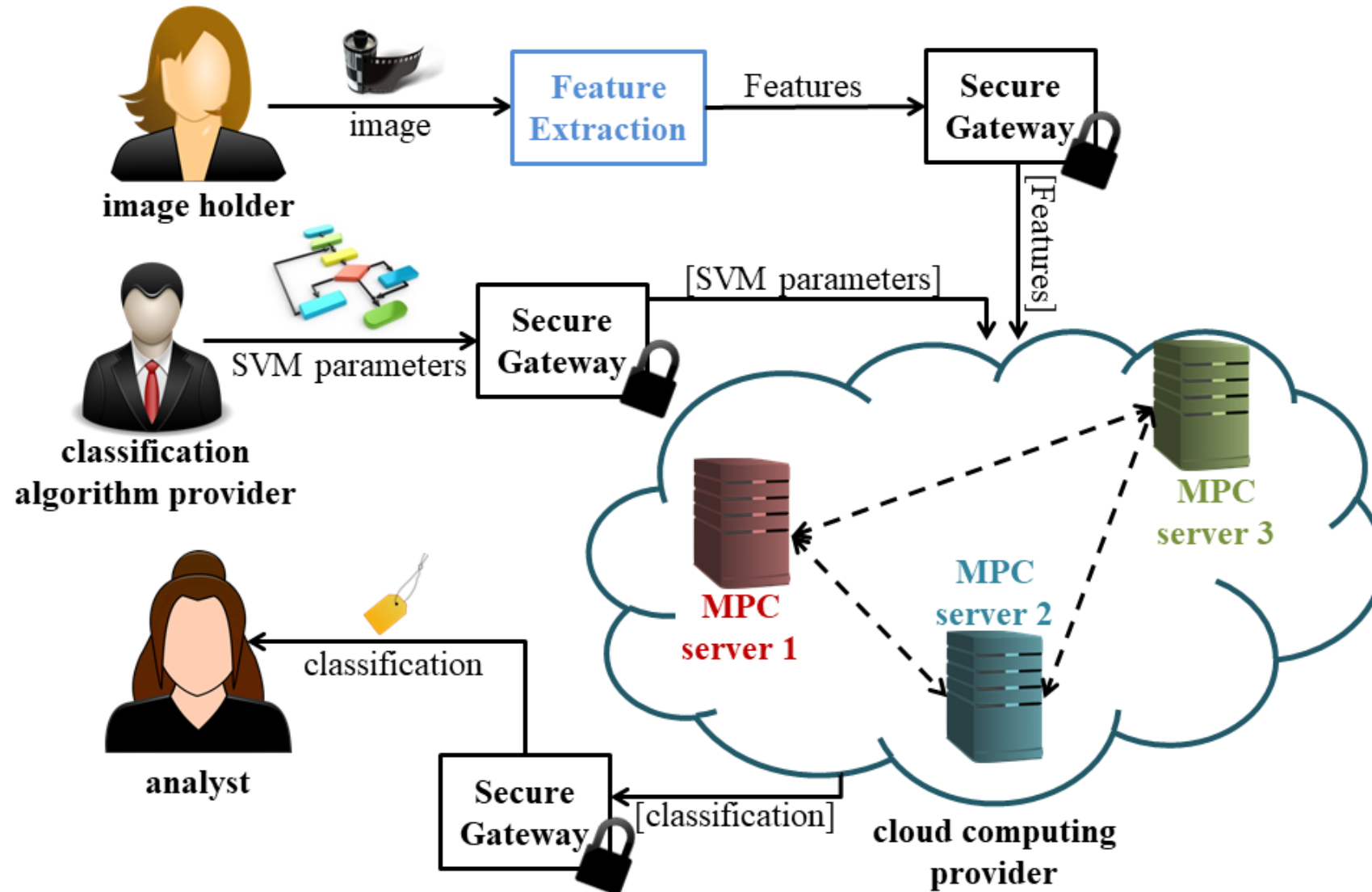
Concrete Application Scenarios with MPC

EPIC: Efficient Private Image Classification

Image Classification



EPIC: Efficient Private Image Classification



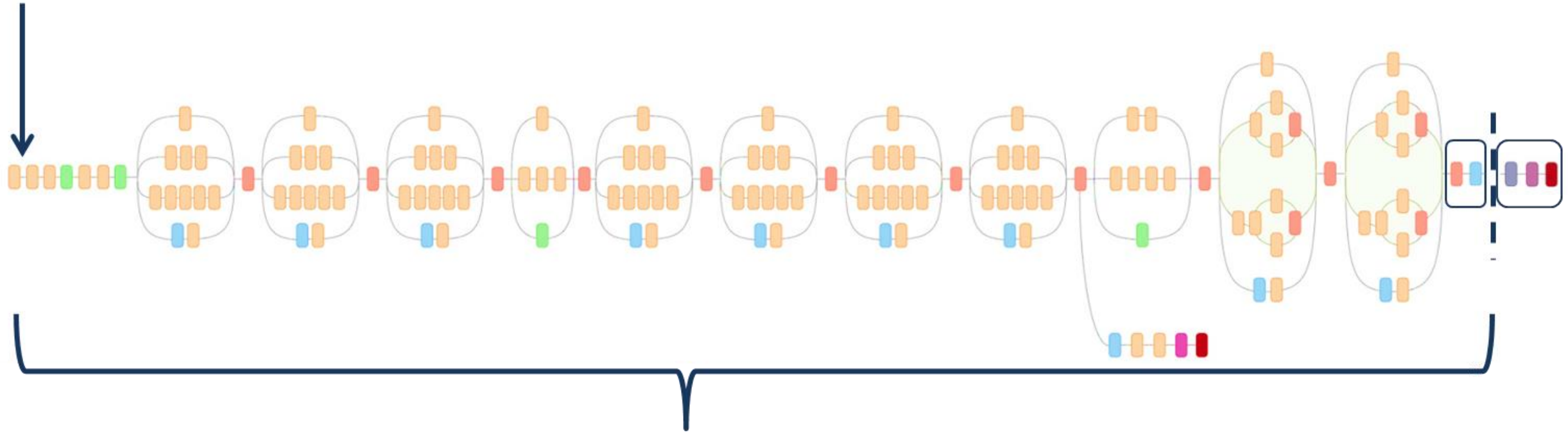
Zoom into Feature Extraction



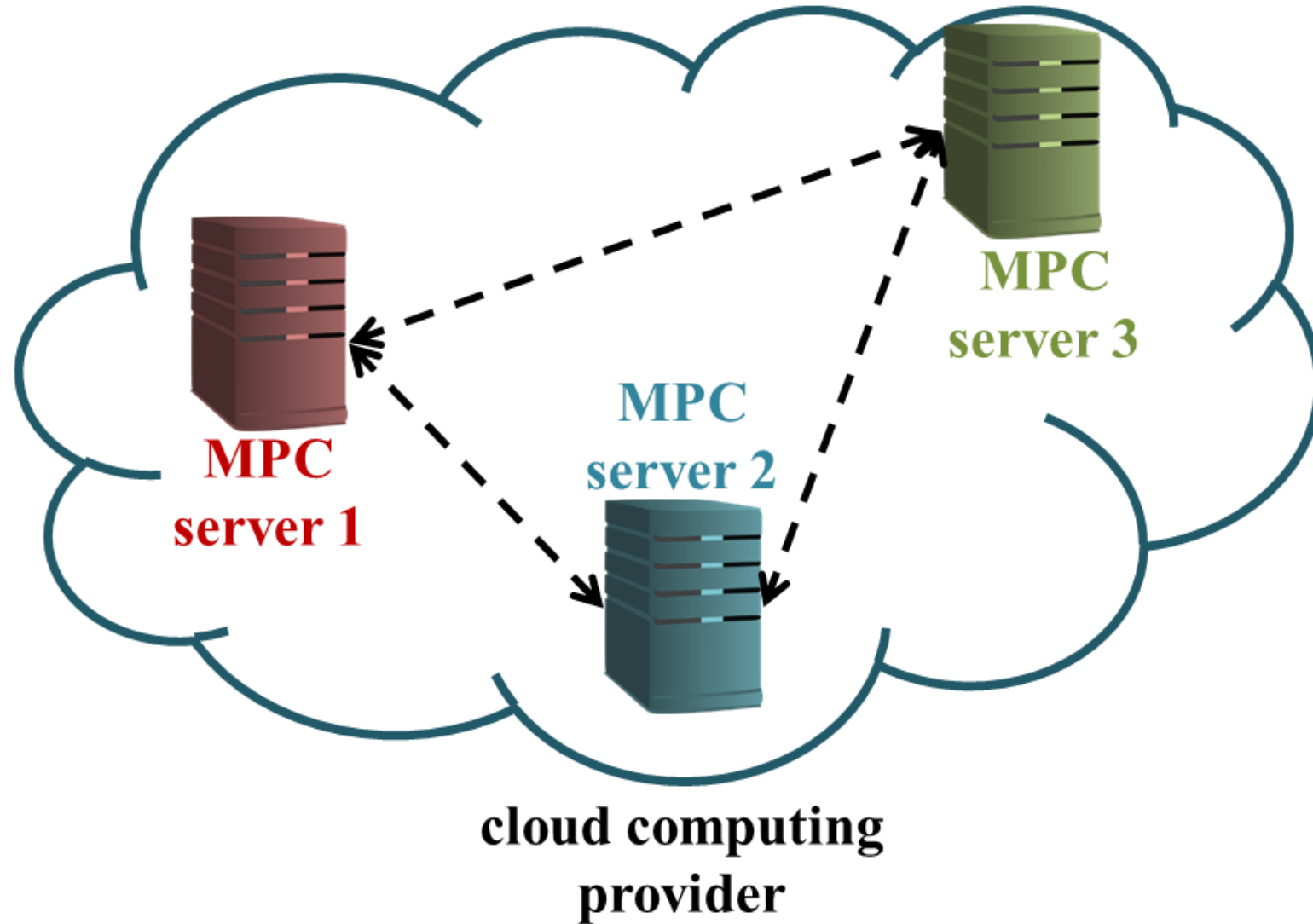
Transfer Learning Feature Extraction (or: Learning from the Masters)



Plaintext (non-sensitive)
images



Where does the MPC happen?



EPIC Security

Active Security

EPIC



vs.

Passive Security

All other protocols in the related work



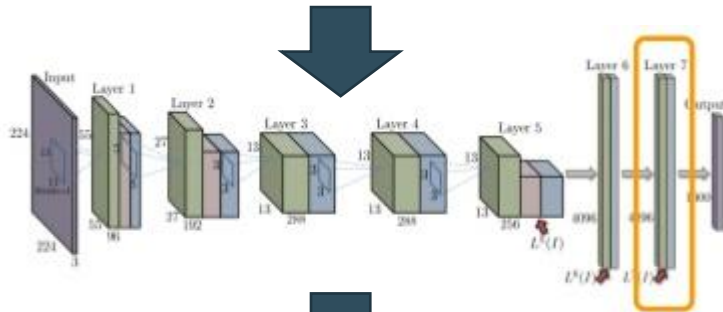
Step 1: Create the ML Model



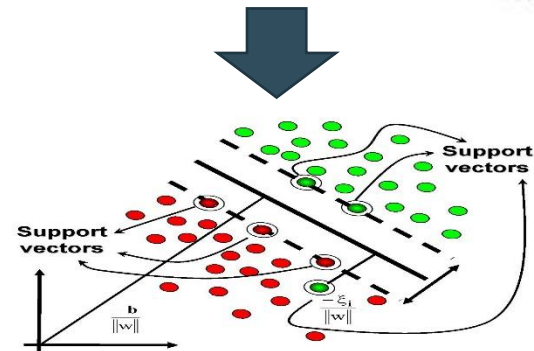
Alice



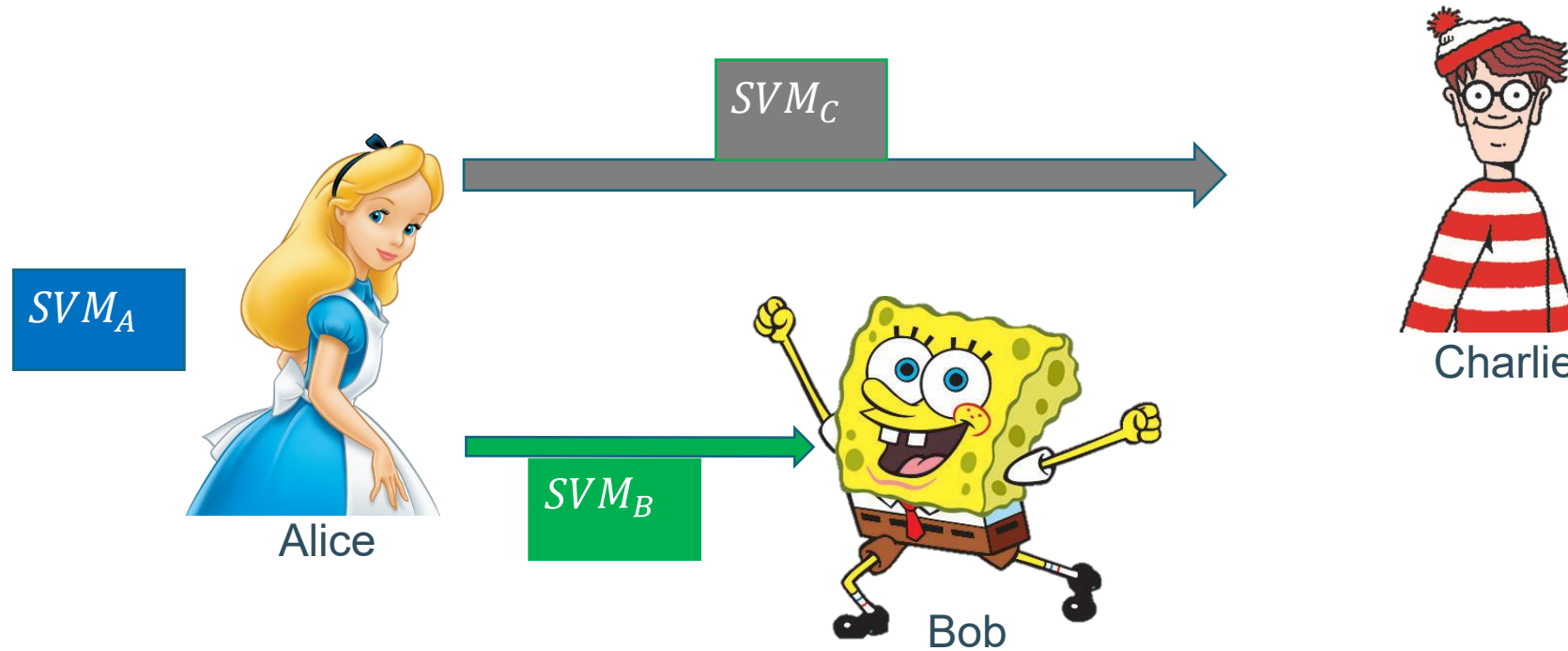
Inception-v3 CNN



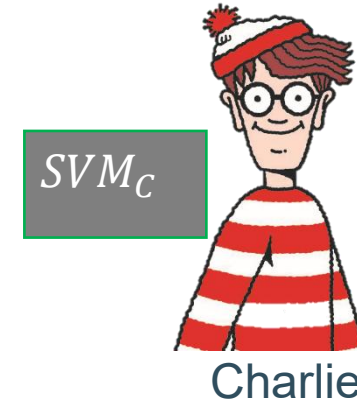
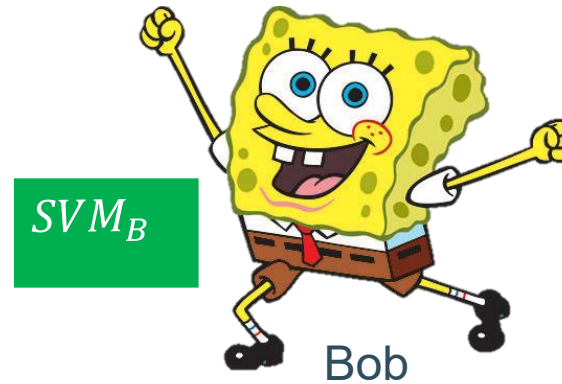
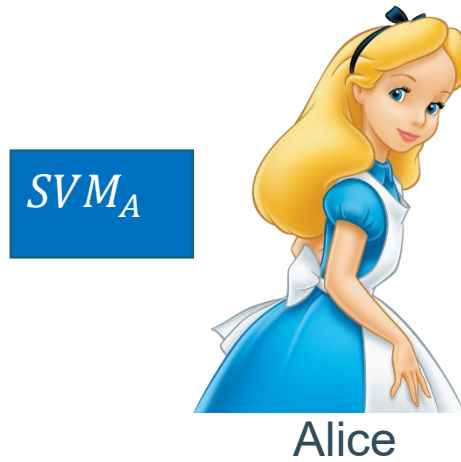
Linear SVM



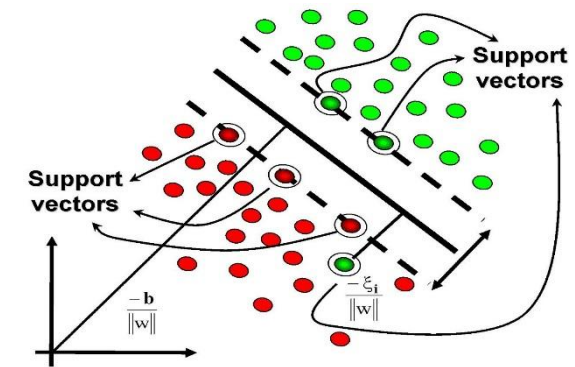
Step 2: Alice secret shares the ML model



Step 2: Alice secret shares the ML model



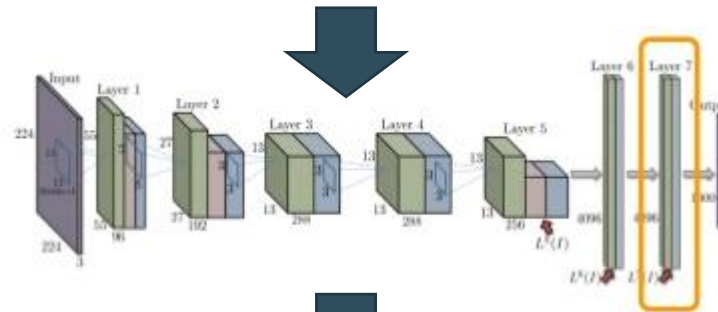
$$SVM_A + SVM_B + SVM_C =$$



Step 3: Bob extracts features



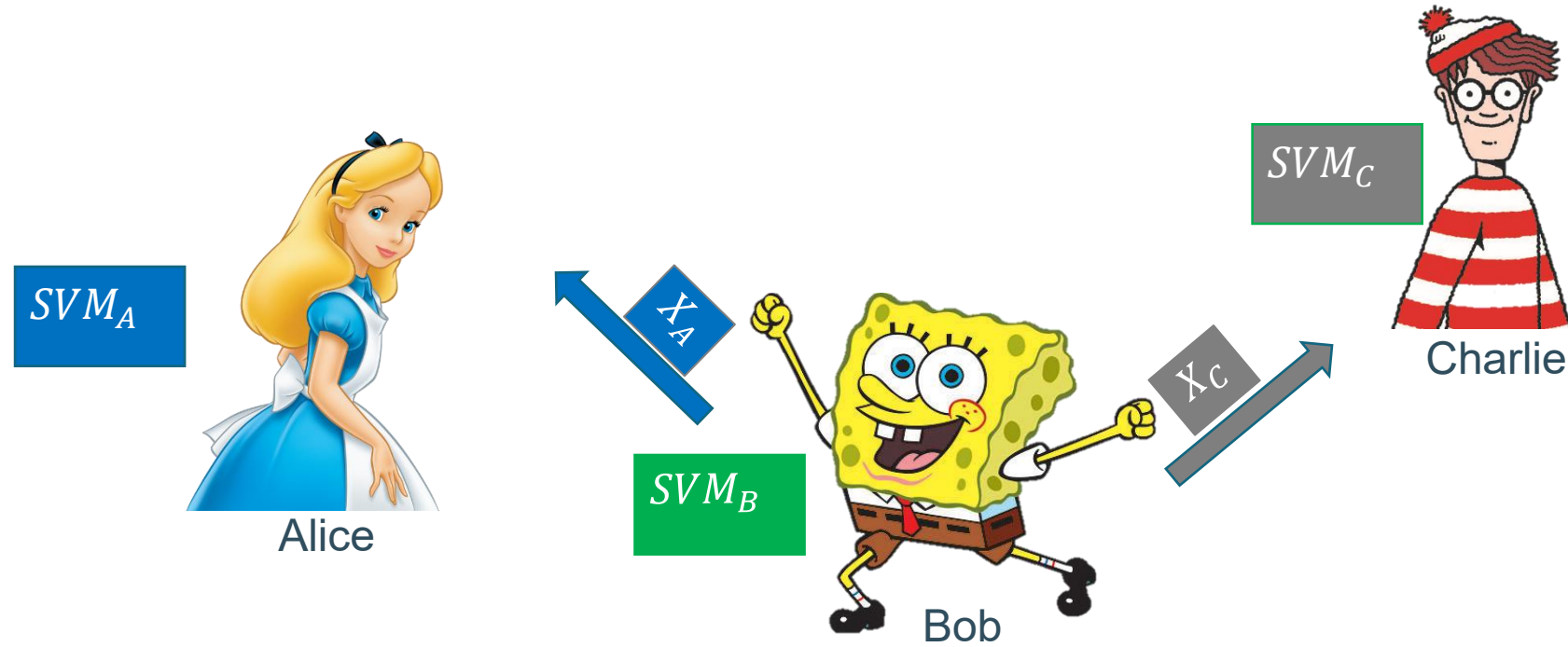
Inception-v3 CNN



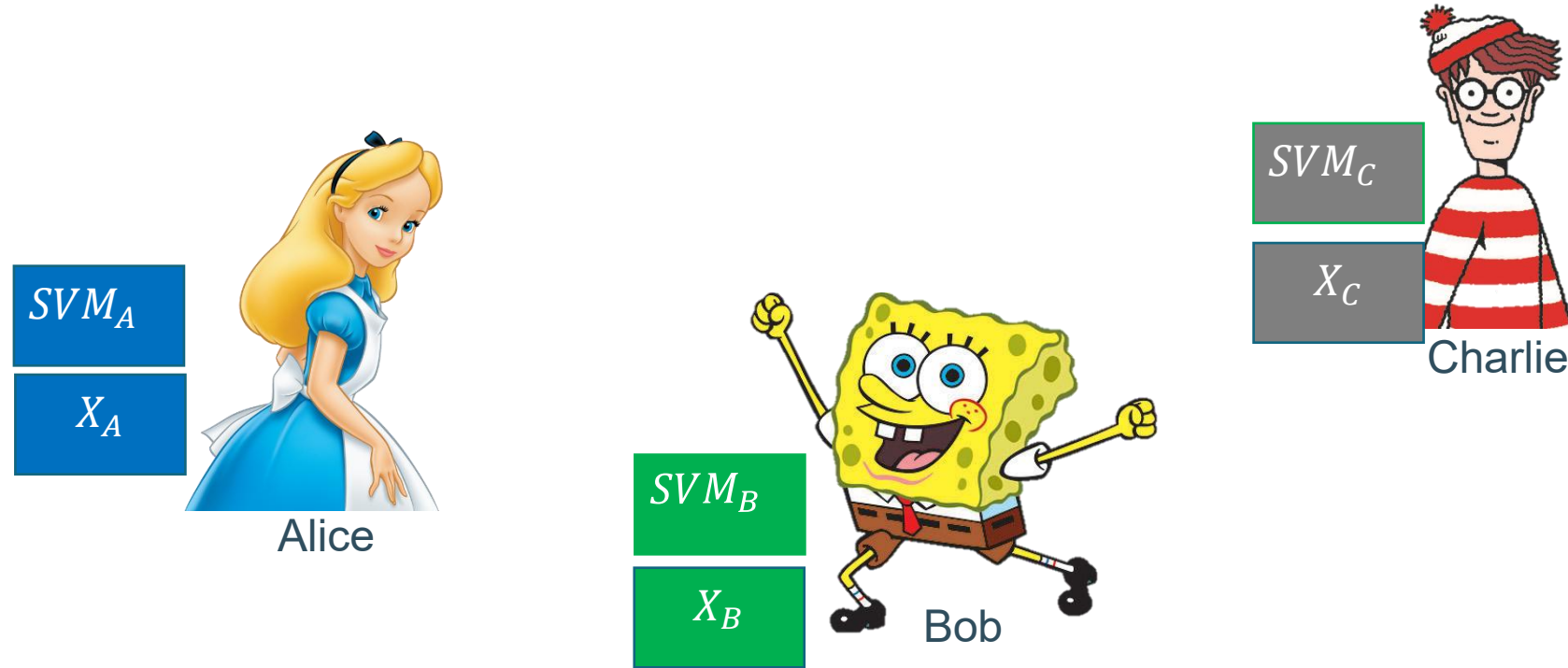
Features

X

Step 4: Bob secret shares features

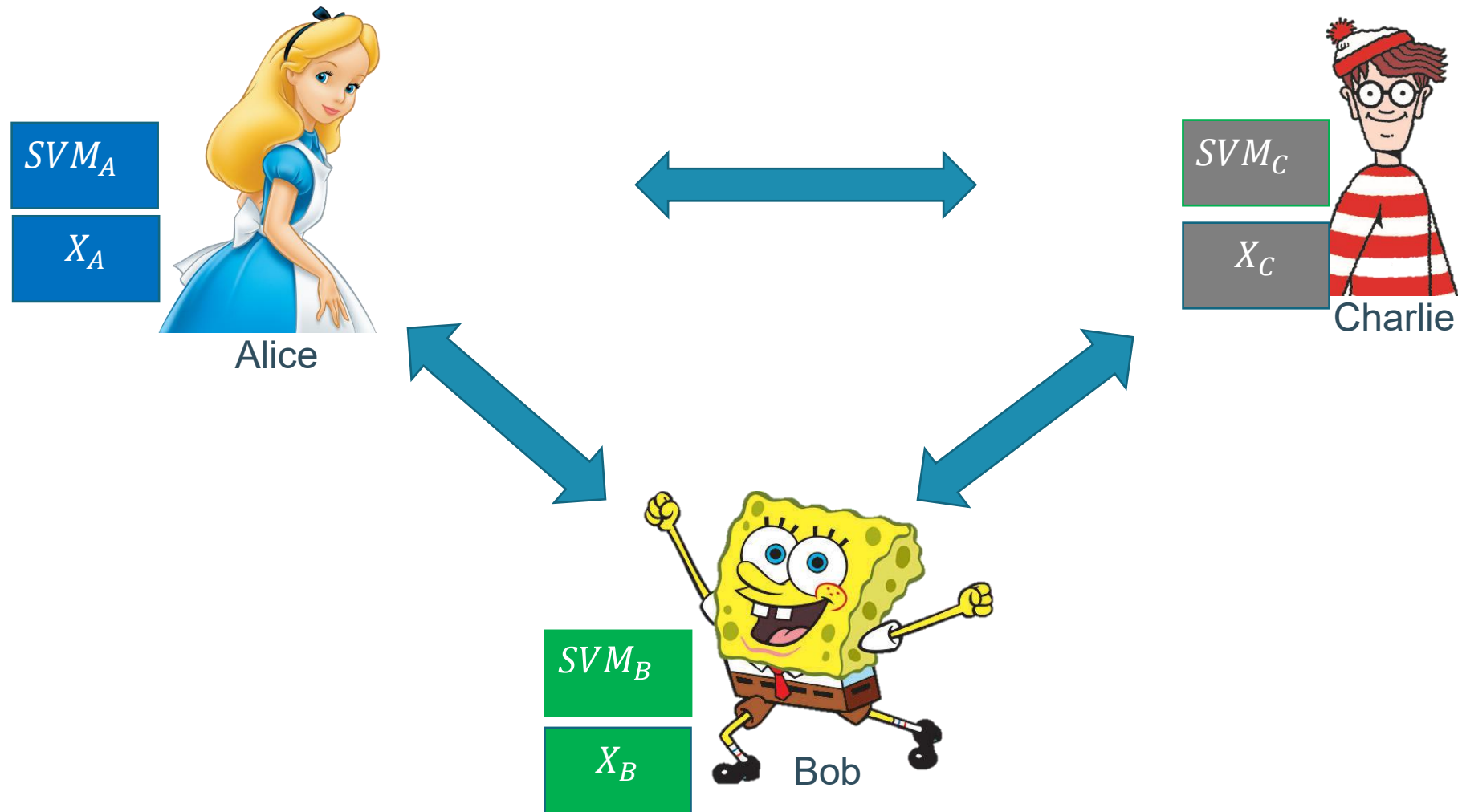


Step 4: Bob secret shares features

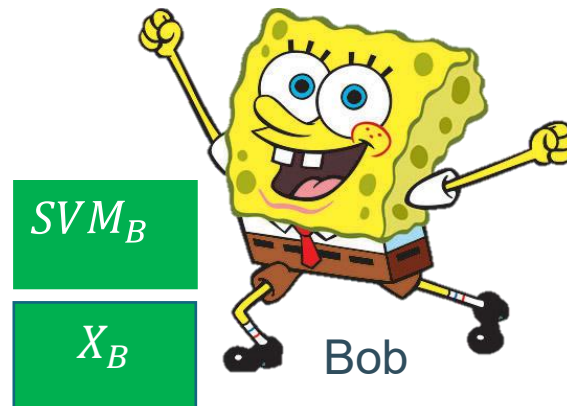
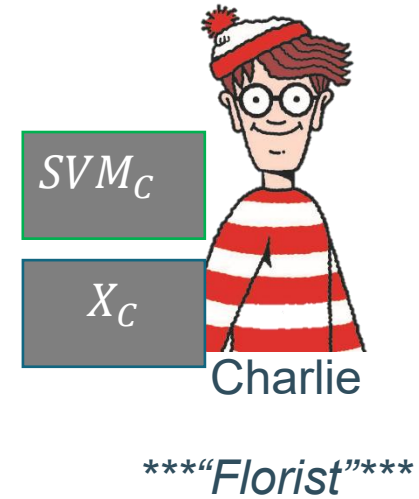
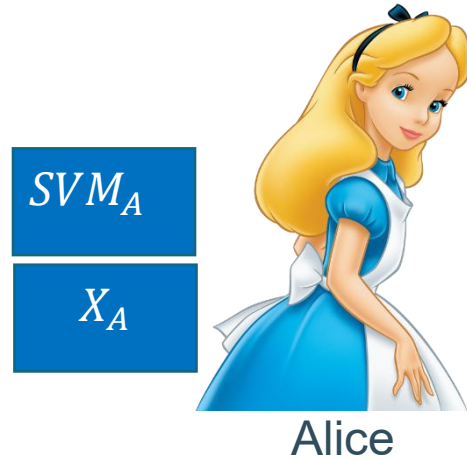


$$X_A + X_B + X_C = \text{CNN-Feat}(\text{Image of a man in an apron})$$

Step 5: Parties use MPC to help Charlie compute label of SVM-Alice(Bob-Image)

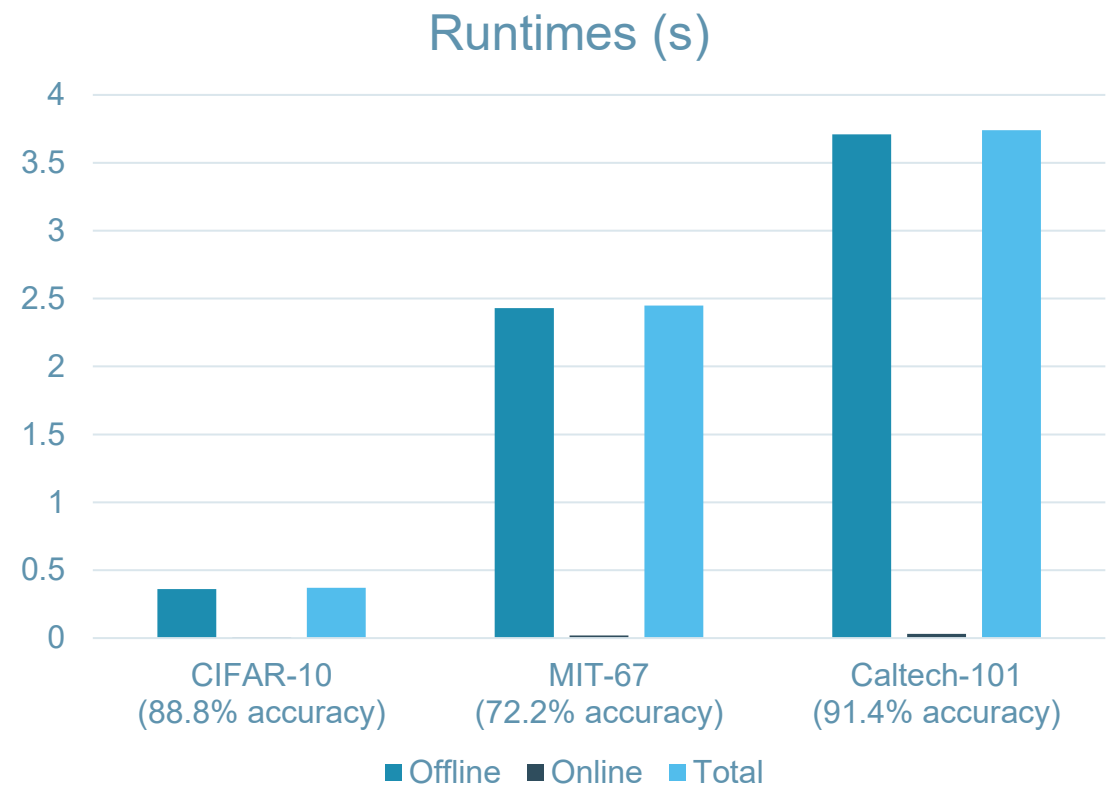


Step 5: Parties use MPC to help Charlie compute label of SVM-Alice(Bob-Image)

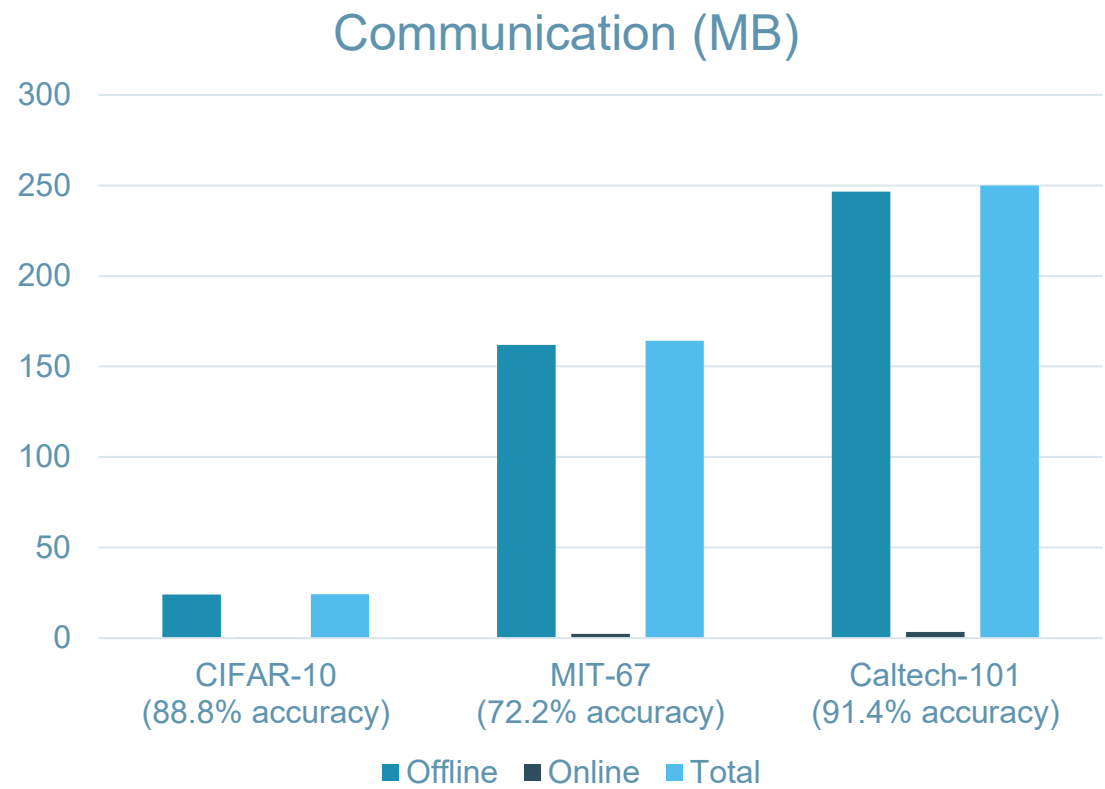


EPIC Performance – Simple Variant

Computation Cost

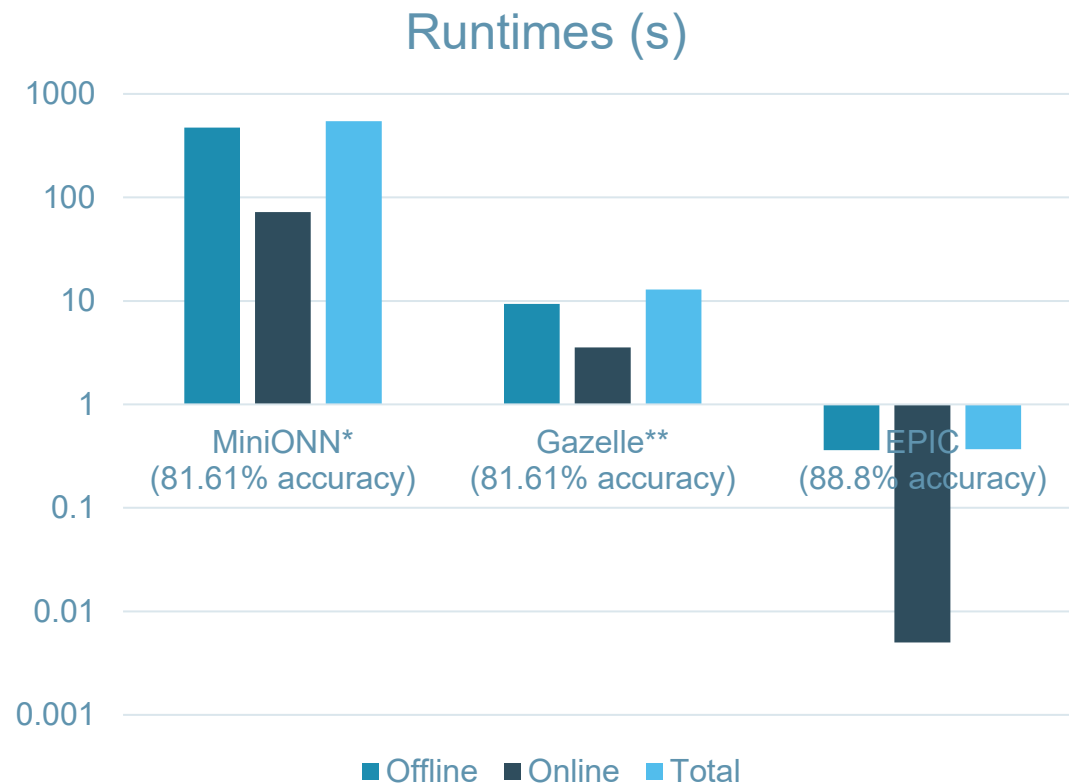


Communication Cost

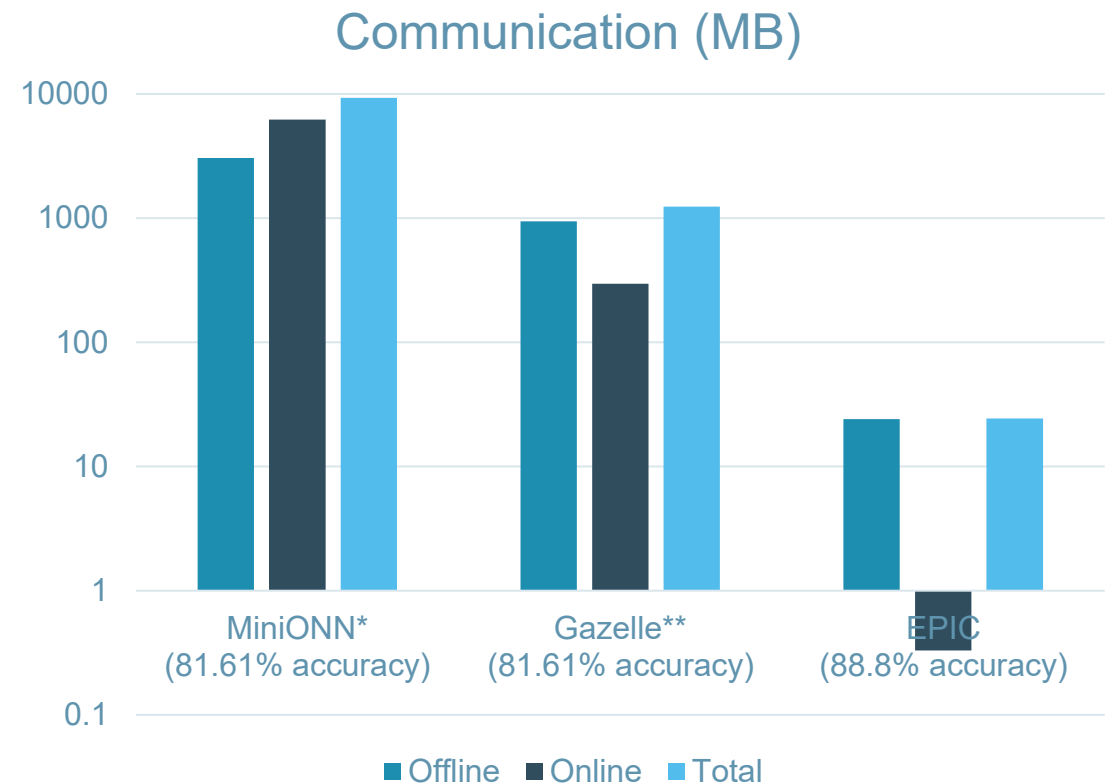


Performance of the state-of-the-art private image classification

Computation Cost



Communication Cost



* Jian Liu, Mika Juuti, Yao Lu, N. Asokan. **Oblivious Neural Network Predictions via MiniONN Transformations**. In *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security* (pp. 619-631). ACM.

** Chiraag Juvekar, Vinod Vaikuntanathan, and Anantha Chandrakasan. **GAZELLE: A low latency framework for secure neural network inference**. In *27th USENIX Security Symposium (USENIX Security '18)*, Baltimore, MD, 2018. USENIX Association.

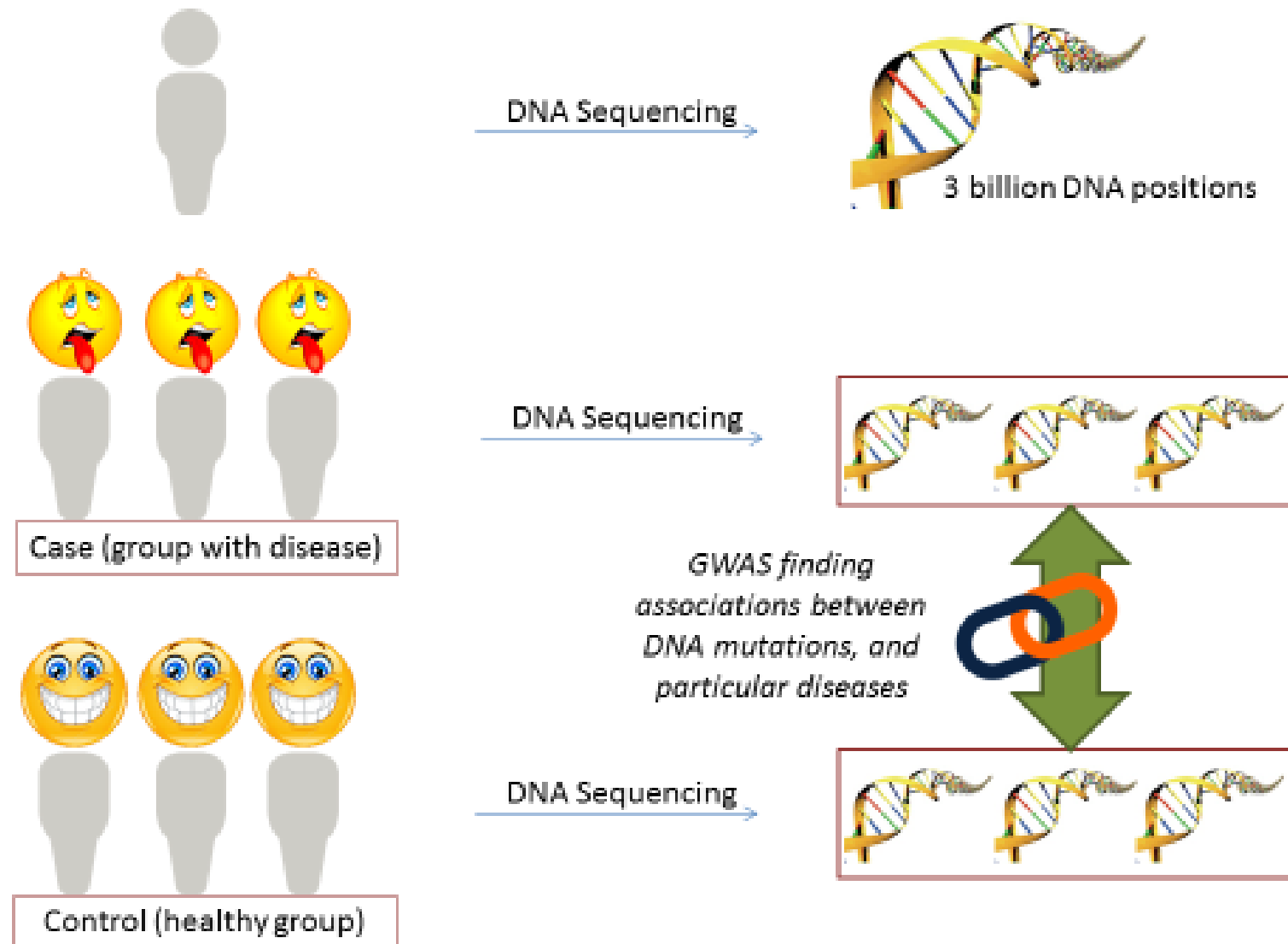
EPIC Efficiency Gain over the state-of-the-art

- EPIC vs. Gazelle¹ on CIFAR-10:
 - 34 times faster runtime;
 - 50 times improvement of communication cost;
 - 7% higher classification accuracy.
- EPIC vs. Gazelle¹ with the same accuracy:
 - 700 times faster runtime;
 - 500 times improvement of communication cost.

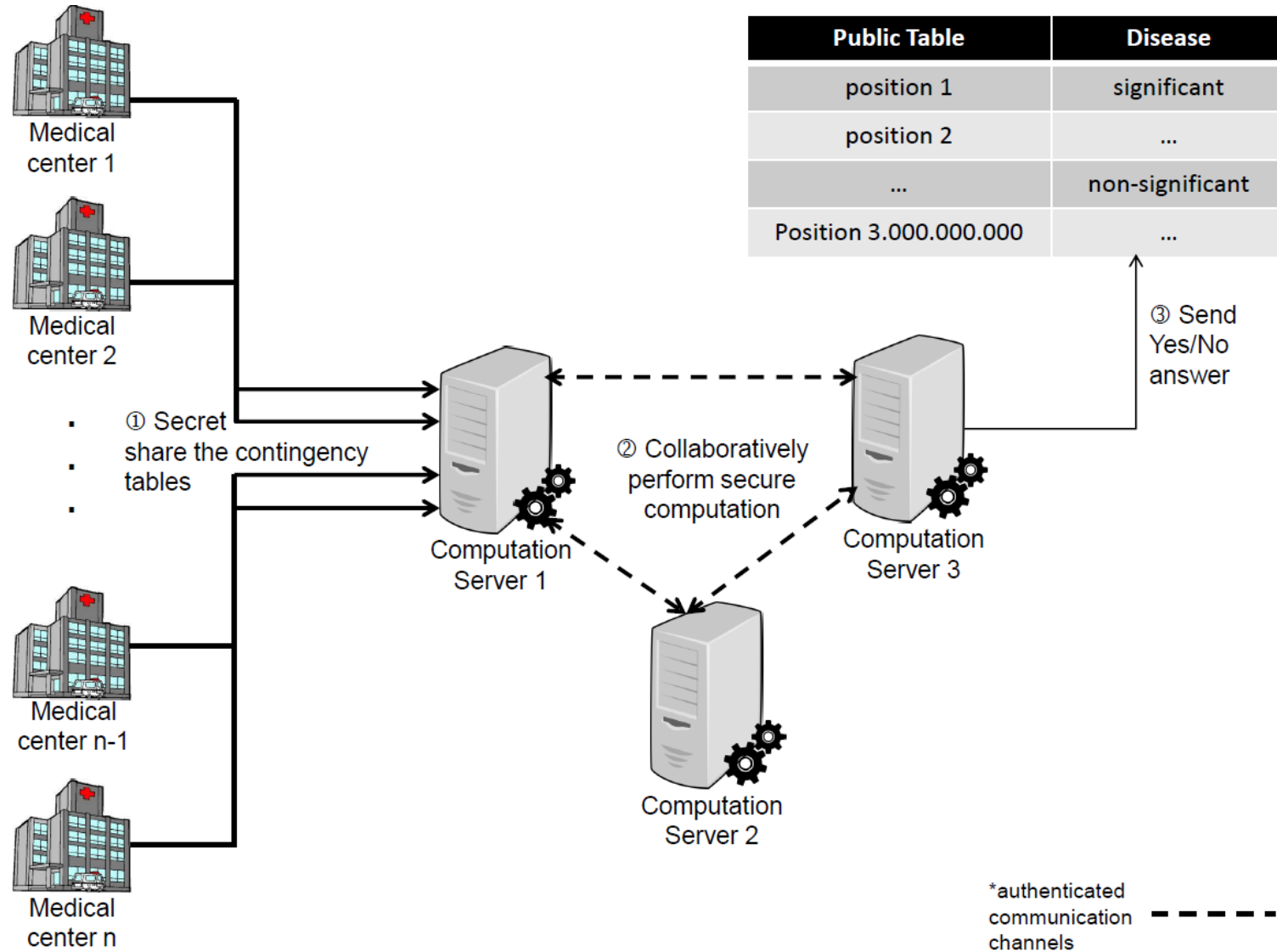
¹ Chiraag Juvekar, Vinod Vaikuntanathan, and Anantha Chandrakasan. **GAZELLE: A low latency framework for secure neural network inference**. In *27th USENIX Security Symposium (USENIX Security '18)*, Baltimore, MD, 2018. USENIX Association.

Privacy Preserving Genome Wide Association Studies

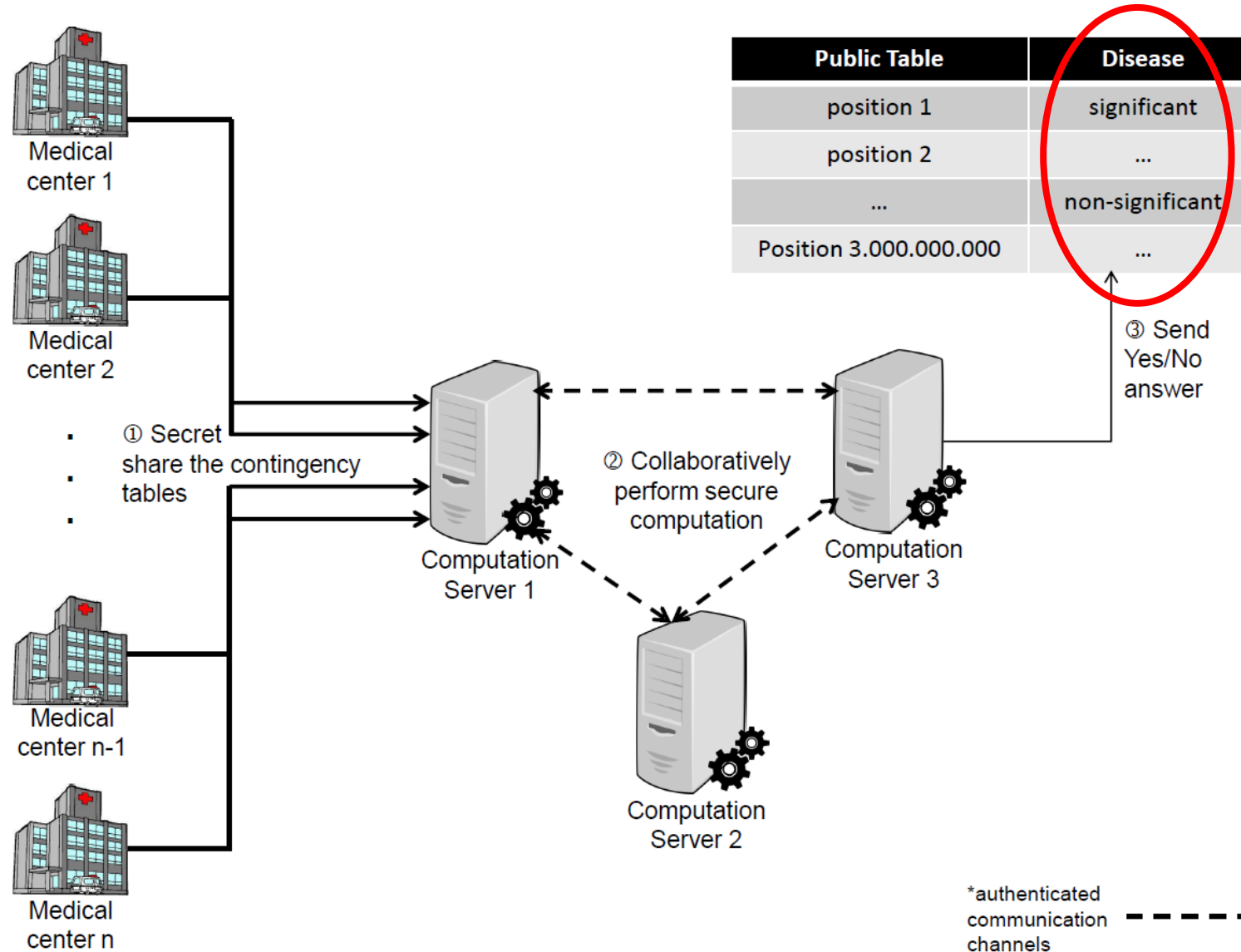
Privacy Preserving GWAS



MPC-based Solution



MPC-based Solution



Problem of re-identification



Problem of re-identification



MPC-based GWAS – Performance

Medical Centers	Number of Patients	CPU Time (Server 1)	Data Sent (Server 1)	CPU Time (Server 2, and Server 3)
20	200000	2.2ms	12.7KB	1.9ms
40	400000	2.3ms	17.8KB	2.0ms
60	600000	2.3ms	23KB	2.0ms
80	800000	2.5ms	28.1KB	2.2ms
100	1000000	2.4ms	33.2KB	2.1ms

MPC-based GWAS

vs.

HE-based GWAS

- CPU time in the range of milliseconds
- Total communication cost in the range of KB
- Multiple parties (≥ 2) required to perform the computation
- Active Security Guarantees

- CPU time in the range of seconds
- Total communication cost in the range of MB
- One party performs the computation
- Semi-honest Decryptor assumption

References

- David Evans, Vladimir Kolesnikov, Mike Rosulek, A Pragmatic Introduction to Secure Multi-Party Computation, Available at:
<https://www.cs.virginia.edu/~evans/pragmaticmpc/pragmaticmpc.pdf>
(Chapters 1, 2.1, 2.2, and 3.1)
- See also handout for application references

Outlook

Prepare to ace the final exam!

Questions?



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Bij ons leer je de wereld kennen